

Cheap Options Are Expensive

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We show that demand pressure from retail investors makes options on low-price stocks relatively expensive—delta-hedged options on low-price stocks underperform those on high-price stocks by 0.63% per week for calls and 0.36% for puts. Natural experiments corroborate this finding: options become more expensive following stock splits, options on mini indices are more expensive than those on main indices, and mini contract options are more expensive than standard options. We attribute our findings to retail investors' preference for skewness and divergence of opinion. Limits to arbitrage and strategic quote setting by market makers contribute to, but do not fully explain, this effect. (*JEL* G13, G14)

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There is a large literature on the role of retail investors in financial markets (see [Barber and Odean 2013](#) for a survey). For example, [Akbas et al. \(2015\)](#) show that retail flows into mutual funds (“dumb money”) exacerbate cross-sectional stock mispricing, while hedge fund flows (“smart money”) attenuate mispricing. Focusing on the option markets, [Gârleanu, Pedersen, and Poteshman \(2009\)](#) demonstrate that demand pressure from end users leads to option mispricing. Building on their insight, we argue that retail investors' trading leads to a specific pattern in options: it affects the relative expensiveness of options written on low-price stocks. We argue that budget constraints make trading in options written on high-price

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stocks prohibitively costly for many retail investors and lead to a concentration of retail trading in options on low-price stocks. We then examine two effects that generate positive retail demand for those options: preference for skewness and divergence of opinion. We provide comprehensive evidence of the pricing effects and these potential mechanisms. We show also that limits to arbitrage and strategic quote setting by option market makers potentially contribute to, but do not fully explain, the relative expensiveness of options on low-price stocks.

We start our empirical analysis by constructing delta-hedged call and put decile portfolios sorted on the underlying stock price. Our sample covers at-the-money (ATM) U.S. equity options on individual stocks during the period from 1996-2020. We find that portfolio returns and alphas from factor models increase in the underlying stock prices. For example, the return on the long-short portfolio that is long in delta-hedged calls on high-price stocks and short in delta-hedged calls on low-price stocks is 0.63% per week. We risk adjust using a factor model that includes both stock factors (Fama and French 2015 factors and the momentum factor) and the option factor from Coval and Shumway (2001). Alpha from this 7-factor model is 0.58% per week. Delta-hedged put portfolios show similar patterns (long-short returns and alphas of 0.36% and 0.34%, respectively).

We test whether this pattern can be explained by the correlation of stock price with other stock characteristics. We double sort the option portfolios on stock price and characteristics such as size, book-to-market, profitability, past stock return, forecast dispersion, and net issuance. These characteristics are shown by Zhan et al. (2022) to be useful for predicting option returns. Although the magnitude of long-short returns is reduced compared to the baseline results, we still find 7-factor alphas in the range of 0.17% to 0.58% (statistically significant in all cases). We also confirm the relation between delta-hedged returns and the underlying price via Fama and MacBeth (1973) (FM) regressions. Even after controlling for additional variables related to option Greeks, the coefficient on (log) stock price is positive and statistically significant for both delta-hedged calls and puts. This effect is also economically large as the returns to delta-hedged call portfolios increase by 0.30% per week when the underlying stock price moves by one standard deviation.

Our hypothesis is predicated on the participation of retail investors in options on low-price stocks. We perform two direct tests to validate this assumption. We obtain data from 2009-2017 from International Securities Exchange (ISE) on daily records of option buy and sell activities of different market participants. Using these data, we first run FM regressions of net option buys from different categories of investors on the underlying stock price. We find that the buying of options (both calls and puts) by professional option traders increases with the underlying price. In contrast, retail option traders' buying is inversely related to the stock price.¹ Second, we focus on net investor demand, measured as net open interest. Gârleanu,

¹ The customer category in ISE data that we use to proxy for retail demand contains some trades of professional traders. The professional category is presumably not tainted by misclassification. Since our results obtain for the customer category but not for the professional category, any contamination of institutional trades in the customer data would bias our results downward, working against our hypothesis.

Pedersen, and Poteshman (2009) suggest that weighting the net demand by option Greeks, such as Gamma and Vega, captures the unhedgeable risks. We find that the expensiveness of options on low-price stocks is indeed related to various measures of net option demand (raw or weighted) from retail investors. We take the totality of this evidence to be supportive of our hypothesis that the mispricing of options that we identify is likely driven by less sophisticated retail investors.

The second predicate of our hypothesis is the budget constraints for retail investors. We analyze three quasi-natural experiments to validate the conjecture that budget constraints are related to option expensiveness. First, we analyze mini options on three individual stocks with available mini contracts: Amazon, Apple, and Google. Mini contracts carry a deliverable of 10 shares of an underlying security, unlike standard contracts of 100 shares. The underlying security is exactly the same for mini and standard options. Therefore, the null hypothesis is that the price of a mini option contract should be exactly one-tenth of the price of the corresponding standard contract. We, however, find that the price differences between standard and mini contracts are economically large and vary between 1.1% (calls on Google) to 5.4% (puts on Amazon). Second, we analyze the effect of stock splits on option prices. The option price should scale proportional to the underlying stock price following a split. We find that, for example, calls (puts) are relatively 13% (9%) more expensive compared to their presplit prices in the three days around the split.² We also find a positive coefficient on the split ratio in a panel regression of option returns around event windows. Third, we analyze mini index options traded on both Nasdaq 100 and S&P 500 mini indices (whose prices are one-tenth of the prices of the original indices). Once again, the null is that the prices of the mini index options should be one-tenth of the prices of the regular index options (as the only difference between the regular and the mini indices is the scaling factor). In contrast, we find that mini index options trade at higher relative prices—1-month ATM calls (puts) are 0.9% (1.3%) more expensive on mini indices than on main indices. The magnitudes of these price differences are small as is to be expected given the high liquidity of these options. Nevertheless, the results are strongly statistically significant.

In the final part of the paper, we investigate the two aforementioned mechanisms to explain our results: preference for skewness and divergence of opinion. Prior literature documents preference for skewness for rational reasons (Arditti 1967; Kraus and Litzenberger 1976; Scott and Horvath 1980; Harvey and Siddique 2000; Mitton and Vorkink 2007; Barberis and Huang 2008) or behavioral preferences for lottery-like assets (Green and Hwang 2009; Kumar 2009; Bali, Cakici, and Whitelaw 2011). Preference for lottery-like stocks might also translate into preference for options on such stocks (Blau, Bowles, and Whitby 2016; Byun and Kim 2016). To test for the effect of skewness preferences, we sort

² There is evidence showing that stock return volatility increases following splits (Ohlson and Penman 1985; Brandt et al. 2010). However, stock splits in our sample are anticipated events, and we analyze forward-looking implied volatility. Over the short window of a few days that we analyze, absent the behavioral biases that we study in this paper, we do not expect changes in the implied volatility of options.

all options independently into quintiles based on the market price of the underlying stock and two proxies for the underlying return skewness, namely, idiosyncratic skewness and the MAX10 variable of [Bali, Cakici, and Whitelaw \(2011\)](#). We find that, in the lowest stock-price quintile, the options in the top underlying skewness quintile underperform those in the bottom skewness quintile. This is consistent with the idea that skewness-seeking retail investors, who are more active in options on low-price stocks, are more likely to buy call options on stocks with high underlying returns skewness as such contracts offer even more positively skewed returns. In contrast, there is virtually no effect of the underlying skewness on option returns in the highest stock-price quintile. This is consistent with the other evidence in our paper that retail investors are less active in options on high-price stocks, and with a reasonable assumption that institutional investors are less likely to have preferences for skewness and lottery-like payoffs.

Second, divergence of opinions coupled with short-sale constraints can potentially generate overpricing of securities ([Miller 1977](#)). We expect a greater degree of mispricing when the quality of information about the firm is low and the investors' opinions are likely more divergent. Because retail investors are most active in options on low-price stocks due to budget constraints, the effect should be contained to those options. We use four proxies for the degree of information uncertainty/divergence of opinion: institutional ownership, idiosyncratic stock return volatility, number of analysts, and analyst forecast dispersion. For all four measures of divergence of opinion, we find that there is a negative effect of divergence of opinion on delta-hedged options returns in the lowest stock-price quintile. For example, for institutional ownership, the spread is 0.31% and 0.30% per week for calls and puts, respectively. Once again, consistent with our hypothesis about the role of retail investors, in the highest stock-price quintile, most of the return spreads are insignificant, and some even have the opposite sign.

Our evidence raises an important question: if options on low-price stocks are relatively expensive, why do sophisticated traders not eliminate this effect through arbitrage? Several factors could impede arbitrage in this setting. First, transaction costs in option markets are substantial, particularly for options on low-price stocks that have wide bid-ask spreads. Our tests show that while transaction costs reduce the magnitude of the effect, it remains significant for effective spreads up to 50% of quoted spreads for calls and puts. Second, market makers can strategically set quotes in illiquid options to protect against pick-off risk, the risk that trades are executed at stale quotes when market makers cannot continuously adjust their quotes to reflect changes in the underlying price. To address this concern, we examine options with bid prices greater than \$0.40, where the potential divergence between quotes and fair values is less severe. Third, the concentration of the effect in less liquid stocks (it largely disappears for S&P 500 stocks) suggests that market liquidity and associated limits to arbitrage play an important role. We perform extensive tests controlling for various dimensions of liquidity through double-sorted portfolios and FM regressions. Although these factors contribute to the persistence of our findings, the effect remains significant

even in settings where arbitrage constraints are less binding and after controlling for multiple liquidity measures.

Our paper is related to three streams of literature. First, our paper contributes to the literature that analyzes mispricing in the options market. The usual objective in these studies is to identify an option- or stock-specific characteristic that signals over- or underpricing in the cross-section of options. Some examples include [An et al. \(2014\)](#); [Bali and Murray \(2013\)](#); [Cao and Han \(2013\)](#); [Goyal and Saretto \(2009\)](#); [Heston et al. \(2023\)](#); [Jones and Shemesh \(2018\)](#); [Ramachandran and Tayal \(2021\)](#); [Vasquez \(2017\)](#); and [Zhan et al. \(2022\)](#). We complement their analyses by proposing and testing two specific mechanism related to trading by budget-constrained retail investors.³

Second, our paper is related to the literature that analyzes the importance of nominal stock prices for asset returns. [Shue and Townsend \(2021\)](#) examine the effect of nonproportional thinking on stock return volatility, betas, and reaction to firm-specific news. [Boyer and Vorkink \(2014\)](#) find evidence of skewness preference in options. [Birru and Wang \(2016\)](#) posit that investors overestimate the skewness of low-price stocks and find that out-of-the-money (OTM) calls (but not OTM puts) on low-price stocks are overpriced. In contrast, we find evidence of overpricing for both calls and puts on low-price stocks. [Blau, Bowles, and Whitby \(2016\)](#) and [Byun and Kim \(2016\)](#) show demand and price effects on calls on lottery stocks. We find evidence of overpricing for both calls and puts, and additionally show that the overpricing related to skewness shows up only in options on low-price stocks. In addition, we pinpoint the role of divergence of opinion. Our study complements these studies by relating overpricing to demand pressure from retail traders and budget constraints.

Finally, our paper is related to studies of demand-based option pricing. The pioneering study of [Gârleanu, Pedersen, and Potesman \(2009\)](#) shows that the inability of risk-averse option market makers to perfectly hedge their inventories has an impact on option prices. [Ni, Pearson, and Potesman \(2005\)](#) and [Golez and Jackwerth \(2012\)](#) show that the rehedgeing of option positions just before expiration produces measurable changes in stock price, and [Ni et al. \(2021\)](#) show that option market maker rebalancing impacts stock return volatility. We complement these studies by showing that the hedging costs of option market makers coupled with trading by budget-constrained retail investors has a first-order impact on option prices.

Hypotheses Development

We argue that option mispricing is likely driven by demand pressure from retail investors. Many retail investors are small and have limited capital. For example, [Bryzgalova, Pavlova, and Sikorskaya \(2023\)](#) document that 46% of retail option

³ [Cao and Han \(2013\)](#) use stock price as a control in their investigation of the relation between option returns and idiosyncratic volatility.

trades are simply one option contract and that 41% of those trades have a dollar value less than \$250. It is difficult for budget-constrained retail investors to trade options on high-price stocks, as these options are relatively expensive. To put things in perspective, consider ATM calls on Nvidia with about 1 month to maturity on March 13, 2024: The price of a single contract (written on 100 shares) is around \$7,400. A single ATM call on Meta with the same maturity would cost around \$2,200. These prices can be prohibitively costly for many retail investors, especially if they wish to hold multiple positions to diversify their portfolios. As a result, retail investors are more likely to turn to options written on low-price stocks as these options have relatively low prices and the investors' budget constraints are less likely to bind.

Hypothesis 1. Because of the budget constraints, retail investors are more active in the trading of options on low-price stocks.

We formally test the role of retail investors and budget constraints in Sections Role of Retail Investors and Budget constraints: Quasi-natural experiments.

It is important to note that, by themselves, budget constraints and the participation of retail investors in the trading of options on low-price stocks do not necessarily lead to relative price differences among options. As long as retail investors are, on average, equally likely to buy and sell those options, the aggregate effect of their trading on option prices is likely to be insignificant. [Gârleanu, Pedersen, and Poteshman \(2009\)](#) show that demand pressure increases option prices. Therefore, in order for options on low-price stocks to be relatively more expensive, some mechanism must create a positive demand pressure from retail investors for such options. We consider two different potential mechanisms that lead to positive retail demand for options: preference for skewness and short-sale constraints (coupled with divergence of opinion).

One channel capable of generating positive retail demand for call options on low-price stocks is preference for skewness. Multiple authors have argued that investors might be attracted to stocks with positive skewness that offer lottery-like payoffs. For example, [Mitton and Vorkink \(2007\)](#) find skewness preferences in individual investor accounts; [Boyer, Mitton, and Vorkink \(2010\)](#) show that expected idiosyncratic skewness is negatively associated with future returns; and [Bali, Cakici, and Whitelaw \(2011\)](#) document underperformance of stocks with high maximum daily returns. Retail investors seeking positively skewed returns can turn to call options thereby inflating the prices of those options. [Birru and Wang \(2016\)](#) argue that investors with a preference for lottery-like stocks inflate the prices of call options on such stocks. We argue that this effect is likely limited to the sample of options on low-price stocks, which are more actively traded by retail investors due to budget constraints. Skewness-seeking retail investors will be particularly attracted to call options on stocks with high return skewness, whose payoffs compound the effect of underlying skewness. However, it is less likely that institutional investors exhibit preference for skewness. We therefore expect the underlying return skewness to play less of a role in

option mispricing among options on high-price stocks. Although this mechanism is relevant for call options, any mispricing of call options is likely to create mispricing of put options in the same direction via the put-call parity. These arguments are summarized in the following hypothesis.

Hypothesis 2a. The relative expensiveness of options on low-price stocks is larger for stocks with more positively skewed returns. There should be a weaker or no effect in options on high-price stocks.

The second channel is related to short-sale constraints. There are multiple hurdles faced by retail investors who wish to short options. First, one needs a margin account with specific trading permissions to be able to write options, and writing naked short and put options usually requires the highest permission level. Second, options are sold on margin, and retail investors are required to deposit a substantial amount to their account to be able to initiate a short option trade. This is problematic for budget-constrained retail investors. In addition, most brokers implement additional margin requirements (on top of those stipulated by exchanges) and many low-price stocks are not marginable (and would require an investor to deposit the full value of the underlying stock to be able to open a short option position). [Miller \(1977\)](#) shows that short-sale constraints coupled with divergence of opinion among traders lead to securities' overpricing.⁴

The asymmetry between positive and negative views leads to positive demand pressure from retail investors for options in low-price stocks and causes these options to become more expensive. The magnitude of overpricing is positively related to divergence of opinion among retail investors—the greater the divergence, the greater the asymmetry between buy and sell demand (due to shorting constraints) and the larger the overpricing. The divergence of opinions is also likely to be stronger for less information-transparent companies. Therefore, in the sample of options on low-price stocks, the relative expensiveness of options is expected to be greater for firms subject to less information transparency. On the other hand, we expect to see a weaker or no effect of information uncertainty on option prices in the sample of options written on high-price stocks, because retail investors are likely less active in those options due to budget constraints. These arguments give rise to the next hypothesis:

Hypothesis 2b. The relative expensiveness of options on low-price stocks is larger for firms subject to less informational transparency. There should be a weaker or no effect in options on high-price stocks.

We note that Hypotheses 2a and 2b are not necessarily mutually exclusive. Indeed, skewness preferences and shorting constraints can both contribute to option mispricing.

Finally, we consider the possibility that market microstructure and liquidity considerations contribute to the relative expensiveness of options on low-price

⁴ See also [Jones and Lamont \(2002\)](#) and [Berkman et al. \(2009\)](#) for empirical evidence.

stocks. Market makers may set their quotes strategically to protect against pick-off risk—the risk of trades being executed at stale quotes when market makers cannot continuously monitor and adjust their positions. Since bid quotes are bounded by zero while ask quotes are potentially unbounded, this asymmetry could result in midpoint quotes exceeding true option values, particularly for less liquid options on low-price stocks. Furthermore, the general illiquidity of these options can create limits to arbitrage that prevent sophisticated traders from eliminating any relative price differences. These arguments lead to our third hypothesis:

Hypothesis 3. The relative expensiveness of options on low-price stocks can be partially driven by strategic quote setting by market makers, illiquidity, and limits to arbitrage.

Option Returns and the Underlying Price

Data and variables

Our primary source of data is IvyDB US OptionMetrics which provides comprehensive coverage of U.S. equity options from 1996–2020. OptionMetrics provides daily closing bid-ask quotes (as well as daily trading volume and open interest) on standard American options. Following standard practice (see, for example, [Zhan et al. 2022](#)), we impose many filters on the option data. We remove options with 0 open interest and options with 0 trading volume because such options are illiquid and their quotes are less likely to reflect any useful information. We eliminate observations that violate arbitrage bounds, for which the ask price is lower than the bid price, or for which the bid price is equal to 0. We retain only options maturing on the third Friday of a month.

We initiate our positions at the end of option expiration Friday and hold these positions until the option expiration on Friday of the next month. We note that our filters apply only to the initial date of the portfolio and, as such, do not suffer from the sample selection issues highlighted by [Duarte et al. \(2024\)](#). For each underlying security in each month, we pick a single call option and a single put option closest to ATM, within moneyness (ratio of stock price to strike price) boundaries of 0.8 and 1.2.⁵ We merge the option data with underlying equity data obtained from the Center for Research in Security Prices (CRSP) data set, using the matching algorithm provided by OptionMetrics. The resulting sample includes 316,325 call options and 248,653 put options. To gauge the effect of the underlying price on option prices, each month we sort all options into 10 equal-sized deciles based on the market price of the underlying stock.

[Table 1](#) presents the summary statistics for our sample. We report statistics for characteristics of the options and their underlying stocks for the full sample, as well as for decile 1 (low stock price) and decile 10 (high stock price). For each

⁵ This filter follows the standard approach in the recent options literature. See, for example, [Duarte et al. \(2024\)](#) and [Zhan et al. \(2022\)](#). We use a relatively wide range of moneyness to ensure that our portfolios are well populated, particularly for options on low-price stocks, due to the granularity of strike prices.

Table 1
Summary statistics

		Calls			Puts		
		All	Decile 1	Decile 10	All	Decile 1	Decile 10
<i>Stock price</i>	Mean	39.90	5.84	116.59	42.22	6.45	123.39
	Median	31.07	5.91	105.29	33.02	6.52	110.23
<i>Option price</i>	Mean	1.78	0.50	4.48	1.93	0.69	4.72
	Median	1.36	0.44	4.05	1.49	0.61	4.21
<i>Size</i>	Mean	11.33	0.99	31.90	13.60	1.29	36.25
	Median	2.63	0.37	15.27	3.25	0.47	17.75
<i>Market-to-book</i>	Mean	4.32	2.98	7.07	4.51	3.02	7.40
	Median	2.85	1.71	4.84	2.93	1.69	5.10
<i>Past stock return</i>	Mean	0.099	-0.043	0.204	0.092	-0.049	0.207
	Median	0.055	-0.127	0.159	0.046	-0.136	0.162
<i>Profitability</i>	Mean	0.079	-0.231	0.215	0.092	-0.205	0.219
	Median	0.132	-0.121	0.205	0.140	-0.095	0.209
<i>Analyst forecast dispersion</i>	Mean	0.074	0.030	0.060	0.075	0.038	0.061
	Median	0.058	0.057	0.042	0.058	0.065	0.043
<i>Net shares issuance</i>	Mean	3.92	9.86	1.76	3.76	9.71	1.76
	Median	0.97	3.51	0.28	0.92	3.47	0.30
<i>Stock illiquidity</i>	Mean	0.368	1.538	0.024	0.263	1.086	0.020
	Median	0.088	0.925	0.013	0.063	0.621	0.011
<i>IVOL</i>	Mean	0.024	0.040	0.017	0.024	0.040	0.017
	Median	0.022	0.038	0.015	0.021	0.037	0.015
<i>ISKEW</i>	Mean	0.232	0.306	0.304	0.188	0.253	0.285
	Median	0.216	0.372	0.201	0.187	0.341	0.190
<i>IKURT</i>	Mean	4.436	4.915	3.904	4.441	5.047	3.898
	Median	1.977	2.444	1.716	1.969	2.518	1.700
<i>MAX10</i>	Mean	0.046	0.070	0.034	0.045	0.069	0.035
	Median	0.041	0.066	0.031	0.040	0.065	0.032
<i>Option volume</i>	Mean	287.1	255.6	395.2	228.1	180.4	301.7
	Median	43.3	47.4	90.5	35.4	32.3	66.5
<i>Open interest</i>	Mean	1,765.2	2,032.3	1,936.4	1,333.1	1,553.7	1,279.6
	Median	442.6	556.1	627.2	324.6	373.9	403.5

(continued)

Table 1
Continued

		Calls			Puts		
		All	Decile 1	Decile 10	All	Decile 1	Decile 10
<i>Option volume/open interest</i>	Mean	0.482	0.429	0.452	0.578	0.513	0.573
	Median	0.138	0.116	0.174	0.166	0.138	0.211
<i>Option bid-ask spread</i>	Mean	0.192	0.363	0.095	0.166	0.290	0.090
	Median	0.133	0.294	0.076	0.119	0.231	0.072
<i>Implied volatility</i>	Mean	0.482	0.801	0.339	0.486	0.806	0.347
	Median	0.437	0.777	0.317	0.440	0.773	0.324
<i>IV – HV</i>	Mean	–0.005	0.039	–0.013	0.003	0.047	–0.008
	Median	–0.007	0.034	–0.008	–0.001	0.036	–0.005
<i>Gamma</i>	Mean	0.145	0.350	0.051	0.135	0.322	0.047
	Median	0.114	0.326	0.048	0.106	0.299	0.045
<i>Vega</i>	Mean	4.413	0.617	13.128	4.670	0.686	13.886
	Median	3.373	0.622	11.855	3.581	0.687	12.408

The data on options are from the IvyDB U.S. OptionMetrics database over the period 1996–2020. Data on the underlying stocks are obtained from CRSP/Compustat. The sample includes data on all options as of the option expiration day in each month, the third Friday of the month. The selected options are those that expire in the following month. For each underlying security in each month, we pick the call and the put options that are closest to ATM, as long as the moneyness (ratio of *Stock price* to strike price) is between 0.8 and 1.2. We exclude options with 0 open interest or trading volume. The final sample includes 316,325 call options and 248,653 put options. Each month we sort all options into 10 equal-sized deciles according to the market price of the underlying stock. The table reports statistics of the characteristics of the options as well as their underlying stocks for the full sample and separately for the top and bottom deciles (decile 1 = lowest stock prices, decile 10 = highest stock prices). All variables are winsorized at the 1st and 99th percentiles. For each variable, we first calculate the cross-sectional mean and median across stocks for each portfolio. We then report the time-series averages of these means/medians. *Size* is the equity value of the underlying stock (in billions of dollars). *Market-to-book* of the underlying stock is the ratio of current equity market value to equity book value as of the previous quarter. *Past stock return* of the underlying stock is cumulative return over the past six months. *Profitability* is the annual income before extraordinary items divided by the previous year's book equity value. *Analyst forecast dispersion* is the ratio of the standard deviation of analyst earnings forecasts to the absolute value of the consensus mean forecast; for each firm-month, we use the average dispersion over the previous three months. *Net shares issuance* (in percent) is measured by the annual log change in split-adjusted shares outstanding. *Stock illiquidity* of the underlying stock is Anilind's (2002) measure, calculated by the monthly average of daily ratios of absolute return to dollar trading volume (in millions). *IVOL*, *ISKEW*, and *IKURT* are the standard deviation, skewness, and kurtosis, respectively, of the residuals from regressions of daily stock returns on the Fama–French (1993) three factors in the past three months, and *MAX10* is the average of the highest 10 daily returns during that period. *Option volume/open interest* is the ratio of the daily trading volume to the open interest of the option. *Option bid-ask spread* is the difference between the ask and the bid quotes of the option over the midpoint of bid and ask quotes at the end of the trading day. Implied volatilities are provided by OptionMetrics. *IV-HV* is the difference between the option's implied volatility and historical volatility, based on daily returns over the past year. *Gamma* and *Vega* are Black–Scholes option derivatives.

variable, we first calculate the cross-sectional mean and median across stocks. We then report the time-series averages of these means and medians. To reduce the influence of outliers in our tests, we winsorize all variables at the 1st and the 99th percentiles.

Existing studies (see, for example, [Goyal and Saretto 2009](#); [Cao and Han 2013](#); [Zhan et al. 2022](#)) have identified multiple stock and option characteristics that affect option returns. Following these studies, the first set of characteristics is related to underlying stocks. Size is the equity value of the underlying stock (in millions of dollars). Market-to-book is the ratio of the current equity market value to the equity book value in the previous quarter. Past return of the underlying stock is the cumulative return over the past 6 months. Profitability is the annual income before extraordinary items divided by the previous year's book equity value. Analyst forecast dispersion is the ratio of the standard deviation of analyst earnings forecast to the absolute value of the consensus mean forecast; for each firm-month we use the average dispersion over the previous 3 months. Net shares issuance is measured by the annual log change in split-adjusted shares outstanding. Illiquidity of the underlying stock is the [Amihud's \(2002\)](#) measure, calculated by the monthly average of the daily ratios of absolute return to dollar trading volume (in millions). *IVOL*, *ISKEW*, and *IKURT* are the standard deviation, skewness, and kurtosis, respectively, of the residuals from the regressions of daily stock returns on the [Fama and French \(1993\)](#) three factors in the past 3 months. *MAX10* is the average of the highest 10 daily returns during that period. [Byun and Kim \(2016\)](#) use *ISKEW* and *MAX10* to proxy for lottery characteristics of stocks.

The remaining variables in [Table 1](#) relate to the options. *Volume/open interest* is the ratio of the daily option trading volume of a given option contract to the total open interest for the same contract (at the end of the trading day). This is a measure of the liquidity of a given option contract. The second option liquidity measure that we use is the *Bid-ask spread*, computed as the difference between the ask and the bid quotes at the closing scaled by the midpoint quote. *IV-HV* is the difference between the option's implied volatility (IV) and historical volatility (HV). IVs are provided by OptionMetrics. We compute HVs based on daily returns over the past year. *Gamma* is the second derivative of the option price with respect to the stock price, and *Vega* is the derivative of the option price with respect to volatility. We obtain *Gamma* and *Vega* from OptionMetrics.

By construction, there is a big spread in the underlying prices in the decile portfolios, from an average of \$5.84 in decile 1 to \$116.59 in decile 10. Not surprisingly, decile-1 options (both puts and calls) have lower prices than decile-10 options. Stocks in the first decile also have lower market capitalization, lower market-to-book ratios, lower past returns, and are less profitable than stocks in the tenth decile. Decile-1 stocks also tend to have higher idiosyncratic volatility and more positively skewed returns than decile-10 stocks.

The option characteristics also differ between deciles. Both puts and calls in decile 1 have higher *Gammas*, lower *Vegas*, and higher *IVs* than those in decile 10. Interestingly, the options in decile 1 have higher *IV-HV*, suggesting that these

options are relatively more expensive than those in decile 10. The large differences in both stock and option characteristics across the extreme deciles makes it important to control for those characteristics in our tests. We do so via double-sorts and FM regressions presented later in this section.

Delta-hedged option portfolio returns

We start our analysis by constructing delta-hedged call and put portfolios sorted on the underlying price. We hold the delta-hedged portfolios until the maturity of the options. We closely follow [Cao and Han \(2013\)](#) and [Goyal and Saretto \(2009\)](#) to calculate option returns. The return is equal to the total dollar gain at expiration scaled by the absolute value of the total cost of constructing portfolios at the formation date. Thus, for the delta-hedged calls we scale by $|C_0 - \Delta_{C,0}S_0|$ while for the delta-hedged puts the scaling factor equals $P_0 - \Delta_{P,0}S_0$ (note that the delta of a put option is negative), where S_0 , C_0 , and P_0 are the prices of the stock, call, and put options, respectively, at the initiation of the position and $\Delta_{\cdot,0}$ are the deltas at the initiation of the position. We compute option returns from quote midpoints and use the option deltas provided by OptionMetrics.

[Table 2](#) shows the returns to the equally weighted decile portfolios sorted on the underlying stock price (as in [Table 1](#)). We report mean excess returns (in excess of the risk-free rate) as well as alphas from a factor model. Our factor model includes both stock-based and option-based return factors. The stock factors are the five factors of [Fama and French \(2015\)](#) and the momentum factor. The option factor is the zero-beta S&P 500 straddle factor of [Coval and Shumway \(2001\)](#). These option factors are also used in [Cao and Han \(2013\)](#) and [Zhan et al. \(2022\)](#). We match the holding period of option returns to that for factors. Since some months have 4 weeks between expirations and some 5, we convert all returns to weekly returns (by simply dividing the return by the number of weeks in the holding period, as in [Eisdorfer, Sadka, and Zhdanov \(2022\)](#)). We then report all returns and alphas in percent per week. The 10–1 portfolio is constructed as long in decile 10 (long option and short Δ shares) and short in decile 1 (short option and long Δ shares). For robustness, we also show value-weighted (opt-vw) returns and alphas of the 10–1 portfolio, where value weighting is done according to the market value of the options' open interest.

[Table 2](#) shows that all portfolio returns and alphas of the 7-factor model are negative. This is consistent with the findings of [Bakshi and Kapadia \(2003\)](#), [Cao and Han \(2013\)](#), and [Goyal and Saretto \(2009\)](#). For our purposes, we find that for delta-hedged call and put portfolios, returns and alphas, generally increase (albeit not monotonically) from decile 1 (low underlying price) to decile 10 (high underlying price). For example, the average excess return of delta-hedged calls in the decile-1 portfolio (lowest underlying prices) is -0.75% per week, while the return of the decile-10 portfolio (highest underlying price) is -0.12% per week. The corresponding 7-factor alphas are -0.67% per week for decile 1 and -0.09% for decile 10. Returns and alphas on the delta-hedged put portfolios exhibit a similar

Table 2
Delta-hedged option portfolio returns sorted on underlying stock price

	1	2	3	4	5	6	7	8	9	10	10 – 1	10 – 1 opt-vw
<i>A. Delta-hedged calls</i>												
<i>Mean excess return</i>	-0.75 (-10.78)	-0.34 (-5.69)	-0.23 (-4.24)	-0.15 (-2.74)	-0.11 (-1.87)	-0.13 (-2.68)	-0.13 (-2.54)	-0.12 (-2.33)	-0.13 (-3.28)	-0.12 (-2.99)	0.63 (11.02)	0.48 (5.57)
<i>7-factor alpha</i>	-0.67 (-12.59) [0.48]	-0.26 (-6.51) [0.60]	-0.16 (-4.82) [0.66]	-0.10 (-2.82) [0.63]	-0.06 (-1.32) [0.51]	-0.09 (-2.86) [0.60]	-0.06 (-2.03) [0.68]	-0.07 (-1.94) [0.50]	-0.10 (-4.16) [0.72]	-0.09 (-3.39) [0.62]	0.58 (10.69) [0.20]	0.43 (4.88) [0.08]
<i>B. Delta-hedged puts</i>												
<i>Mean excess return</i>	-0.43 (-8.43)	-0.21 (-4.26)	-0.18 (-3.67)	-0.10 (-2.15)	-0.04 (-0.69)	-0.10 (-2.25)	-0.10 (-2.43)	-0.11 (-2.78)	-0.10 (-2.67)	-0.07 (-1.87)	0.36 (8.84)	0.34 (4.76)
<i>7-factor alpha</i>	-0.39 (-9.95) [0.48]	-0.15 (-4.69) [0.62]	-0.11 (-3.89) [0.68]	-0.07 (-2.26) [0.63]	-0.02 (-0.39) [0.35]	-0.06 (-2.33) [0.68]	-0.06 (-2.62) [0.73]	-0.08 (-3.58) [0.72]	-0.07 (-3.19) [0.70]	-0.05 (-1.93) [0.63]	0.34 (8.36) [0.14]	0.32 (4.23) [0.04]

Portfolios are sorted as in Table 1 (decile 1 = lowest stock price, decile 10 = highest stock price). For each option, we calculate the average weekly returns on delta-hedged positions, based on market prices at the option expiration day in a month and the position's payoff at the option maturity day in the next month. For delta-hedged calls, the return is the gain during the investment period scaled by $(P - \Delta P)$, and for delta-hedged puts, the return is the gain during the investment period scaled by $(P - \Delta P)$, where S , C , and P are the stock, call, and put option prices, respectively. The table shows the portfolios' equal-weighted mean excess weekly returns (in excess of the risk-free rate) and alphas from a 7-factor model. The factors are the Fama and French (2015) five factors, the Carhart (1997) momentum factor, and the Coval and Shumway (2001) zero-beta S&P 500 straddle factor. The last column shows value-weighted returns/alphas on the 10 – 1 portfolio, where the weighting is according to the market value of the option's open interest (opt-vw). All returns and alphas are in percent per week, and the corresponding t -statistics are in parentheses. The adj- R^2 of the regression is reported in square brackets below the t -statistics. The sample period is 1996-2020.

relationship with the underlying price. For example, for delta-hedged puts, mean excess returns on the decile-1 portfolio (lowest underlying prices) are -0.43% per week, while such returns on the decile-10 portfolio (highest underlying prices) are -0.07% per week. The $\text{adj-}R^2$ of the regressions, reported in square brackets below the t -statistics, are in the range of 50% - 60% .

The returns and alphas of the long-short 10–1 portfolio are positive and strongly statistically significant. For example, the weekly average return and the 7-factor alpha for delta-hedged calls are 0.63% and 0.58% , respectively. In annualized terms, our strategy generates 10–1 portfolio 7-factor alphas of 29.90% and 17.73% for delta-hedged calls and puts, respectively. The last column of Table 2 shows that the results are also robust to the alternative weighting scheme where options are weighed by the market price of the options' open interest.⁶

Option returns are nonlinear functions of the underlying stock return over discrete intervals. In principle, delta-hedged option returns are invariant to the underlying return only for continuously adjusted deltas (and then only if the pricing model used to compute deltas adequately describes the stock dynamics). Therefore, the use of linear factor models may not be appropriate to calculate risk-adjusted returns. To alleviate this concern, we also perform conditional beta tests similar to those in Goyal and Saretto (2009). Specifically, we run the following time-series regression:

$$R_{pt} = \alpha_p + (\beta_{0p} + \beta'_{1p} \Theta_{pt-1})' F_t + e_{pt}, \quad (1)$$

where Θ_p are option Greeks (Delta, Gamma, and Vega) calculated as the averages of individual options for each decile. Unreported results show that the alphas from these conditional beta models are very similar to those in Table 2. For example, the 7-factor alphas for the 10–1 portfolio of delta-hedged calls and puts are 0.57% and 0.36% , respectively (vs. the base case scenario of 0.58% and 0.34% , respectively).

One potential concern about Table 2 results is that the underlying price might be correlated with some other characteristics of the stock or the option. For example, as shown in Table 1, low-price stocks tend to have higher idiosyncratic volatility, lower market capitalization, lower market-to-book ratios, and lower past returns. We already partly control for the potential effect of these additional variables by reporting alphas from factor models in addition to raw excess returns. To further alleviate this concern, we perform two different types of test in the following two subsections. First, in Section Double-sorted portfolios, we double-sort our option portfolios according to various characteristics of the stock and on the underlying price of the stock. We then check if within each characteristic-sorted portfolio the underlying price has explanatory power for future option returns. Second, in Section FM regressions, we run FM regressions of option returns on the underlying stock price while controlling for various stock and option characteristics.

⁶ We also check for alphas from a 9-factor model where we add the VOL and JUMP factors of Cremers, Halling, and Weinbaum (2015). In unreported results, we find that the alphas and $\text{adj-}R^2$ s from a 9-factor model regression are very similar to those from a 7-factor model. For example, the 9-factor alpha for the 10–1 portfolio is 0.61% and 0.37% for delta-hedged calls and puts, respectively.

Double-sorted portfolios

To account for potential correlation of the underlying stock price with various stock characteristics, we examine returns to double-sorted portfolios. Each month, we first sort all options into quintiles based on a stock characteristic from Table 1. These characteristics include *Size*, *Market-to-book ratio*, *Past stock return*, *Profitability*, *Analyst forecast dispersion*, and *Net equity issuance*. All of these characteristics have been shown by Zhan et al. (2022) to have predictive power for future option returns. To account for skewness preferences, we also include three variables that proxy for skewness, namely, *IVOL*, *ISKEW*, and *MAX10*. Byun and Kim (2016) also use *ISKEW* and *MAX10* to proxy for lottery characteristics of stocks. We also control for stock's *Amihud illiquidity* and option illiquidity as measured by option *Bid-ask spread*, *Option volume*, and option *Open interest*. Finally, we include two option Greeks, namely, *Gamma* and *Vega*.

We further sort options into deciles according to the market price of the underlying stock. This procedure results in 50 characteristic/stock-price portfolios. We calculate the 7-factor alphas of the 10–1 portfolio for each characteristic quintile. In addition, for each stock-price decile, we average returns across the characteristic quintiles, yielding 10 stock-price decile returns. We then compute the 7-factor alpha of the resulting 10–1 portfolio. This procedure controls for correlation between the underlying stock price and various stock characteristics. For example, if the effect of the underlying price on the delta-hedged option returns is primarily driven by the correlation of the price with size, then the returns of the long-short portfolios in this analysis should be indistinguishable from zero because our portfolios are first sorted on size.

Table 3 reports the 7-factor alphas of the various 10–1 portfolios. The alphas reported in column entitled “Avg” in Table 3 are generally lower than those in Table 2, ranging between 0.17% per week (puts) for stock illiquidity control to 0.58% per week (calls) for market-to-book control. This suggests that the portfolio returns in Table 2 can indeed be partly attributed to the fact that some of the stock characteristics known to predict option returns are correlated with the underlying price. For example, size is clearly positively correlated with stock price and has been found to positively affect option returns (Zhan et al. 2022). Consequently, we find that alphas from size control are smaller (0.39% for calls and 0.19% for puts) than those in the baseline results.

Importantly for us, the effect of the underlying stock price remains highly significant for most stock and option characteristics when performing the double sort.⁷ We conclude that the effect of the underlying stock price on returns to delta-hedged call and put portfolios is robust to the double-sorting procedure.

⁷ A notable exception emerges in the double sort involving option Vega, where the effect is limited to the lowest Vega quintiles for both calls and puts. However, option Vega is mechanically related to the underlying stock price, making it difficult to interpret double-sort results.

Table 3
Delta-hedged option portfolio returns controlling for stock characteristics

Sorting characteristic	Low	Q2	Q3	Q4	High	Avg	Low	Q2	Q3	Q4	High	Avg
	Delta-hedged calls						Delta-hedged puts					
<i>Size</i>	1.37 (8.50)	0.40 (5.25)	0.16 (2.61)	0.02 (0.49)	-0.02 (-0.28)	0.39 (7.22)	0.64 (5.96)	0.20 (3.28)	0.05 (0.83)	0.04 (0.92)	0.02 (0.32)	0.19 (5.05)
<i>Market-to-book</i>	0.58 (7.73)	0.54 (8.41)	0.47 (7.14)	0.54 (8.01)	0.79 (10.65)	0.58 (12.14)	0.35 (5.77)	0.22 (4.49)	0.29 (5.52)	0.31 (5.59)	0.54 (9.10)	0.34 (9.85)
<i>Past stock return</i>	0.84 (9.40)	0.48 (8.06)	0.39 (6.71)	0.30 (5.10)	0.65 (8.91)	0.36 (11.30)	0.49 (7.59)	0.28 (5.61)	0.22 (4.85)	0.15 (3.16)	0.47 (7.40)	0.32 (9.62)
<i>Profitability</i>	0.97 (10.28)	0.37 (5.62)	0.11 (1.88)	0.21 (3.91)	0.43 (6.93)	0.42 (8.71)	0.49 (6.88)	0.21 (3.60)	0.08 (1.60)	0.12 (2.13)	0.31 (5.34)	0.24 (6.43)
<i>Forecast dispersion</i>	0.74 (8.10)	0.05 (0.92)	0.20 (3.58)	0.26 (4.43)	0.57 (7.77)	0.36 (7.74)	0.49 (7.90)	0.03 (0.66)	0.09 (1.82)	0.07 (1.37)	0.25 (4.27)	0.19 (5.14)
<i>Net equity issuance</i>	0.16 (3.22)	0.27 (4.05)	0.43 (5.67)	0.55 (7.11)	0.92 (11.63)	0.47 (9.28)	0.10 (2.47)	0.18 (3.16)	0.18 (2.86)	0.26 (4.02)	0.64 (9.92)	0.27 (6.98)
<i>Idiosyncratic volatility</i>	0.04 (1.07)	0.09 (1.93)	0.17 (2.64)	0.55 (7.22)	0.95 (11.00)	0.36 (7.86)	0.03 (0.97)	0.03 (0.57)	0.03 (0.61)	0.33 (5.14)	0.51 (6.92)	0.19 (4.96)
<i>ISKEW</i>	0.59 (9.17)	0.38 (5.34)	0.50 (7.41)	0.58 (8.04)	0.77 (9.09)	0.56 (10.73)	0.38 (6.99)	0.26 (4.63)	0.20 (3.44)	0.41 (7.02)	0.44 (6.04)	0.34 (8.29)
<i>MAX10</i>	0.02 (0.64)	0.15 (3.53)	0.17 (2.63)	0.54 (7.54)	0.98 (10.66)	0.37 (8.39)	0.03 (0.95)	0.11 (3.05)	0.06 (1.19)	0.23 (3.65)	0.55 (7.20)	0.20 (5.55)
<i>Stock illiquidity</i>	-0.03 (-0.56)	0.09 (1.63)	0.14 (2.31)	0.38 (4.93)	1.32 (8.60)	0.38 (7.12)	0.02 (0.37)	0.08 (1.59)	0.07 (1.34)	0.15 (2.69)	0.52 (6.28)	0.17 (4.46)
<i>Option bid-ask spread</i>	0.26 (3.96)	0.25 (3.50)	0.34 (5.06)	0.52 (7.64)	0.93 (11.78)	0.46 (8.75)	0.15 (2.68)	0.12 (2.33)	0.26 (5.65)	0.28 (4.49)	0.47 (7.07)	0.26 (6.41)
<i>Option bid-ask \$spread</i>	0.67 (8.29)	0.65 (9.19)	0.52 (8.02)	0.61 (8.93)	0.49 (7.20)	0.59 (12.27)	0.34 (5.41)	0.36 (5.38)	0.40 (7.89)	0.40 (6.29)	0.38 (7.13)	0.37 (9.49)
<i>Option volume</i>	0.52 (8.69)	0.57 (9.47)	0.64 (9.06)	0.59 (7.73)	0.56 (7.23)	0.58 (10.98)	0.30 (5.58)	0.37 (6.73)	0.38 (6.33)	0.37 (6.33)	0.30 (4.69)	0.34 (8.44)
<i>Open interest</i>	0.51 (8.60)	0.50 (7.70)	0.59 (7.46)	0.70 (9.46)	0.56 (6.89)	0.57 (10.83)	0.27 (4.49)	0.38 (7.22)	0.40 (6.54)	0.36 (5.85)	0.35 (5.90)	0.35 (8.50)
<i>Gamma</i>	0.36 (5.60)	0.37 (5.18)	0.62 (8.66)	0.88 (11.51)	1.13 (13.32)	0.67 (12.64)	0.41 (7.16)	0.34 (5.88)	0.54 (9.06)	0.59 (9.81)	0.58 (10.15)	0.49 (13.18)
<i>Vega</i>	0.94 (12.39)	0.22 (4.25)	0.03 (0.68)	0.01 (0.26)	0.02 (0.56)	0.24 (9.67)	0.42 (6.92)	0.05 (1.05)	-0.04 (-0.91)	-0.01 (-0.27)	-0.02 (-0.64)	0.08 (3.59)

Each month, we first sort all options into quintiles based on a stock characteristic. The options are then further sorted into deciles according to the market price of the underlying stock, yielding fifty characteristic/stock price portfolios. Delta-hedged option returns are calculated as in Table 2. We present the 7-factor alphas for the 10 – 1 portfolio for each characteristic quintile as well as their averages across the five quintiles. The factors are the Fama and French (2015) five factors, the Carhart (1997) momentum factor, and the Coval and Shumway (2001) zero-beta S&P 500 straddle factor. All alphas are in percent per week, and the corresponding *t*-statistics are in parentheses. The sample period is 1996-2020.

FM regressions

In order to explicitly control for other determinants of the cross-sectional patterns in option returns, we perform a multivariate regression analysis by running FM regressions. An additional advantage of this approach is that it allows us to gauge the marginal influence of the underlying stock price on delta-hedged option returns. The dependent variable is the delta-hedged call or put *Return*. Our explanatory variable of interest is the *Logarithm of the underlying stock price*. According to our hypothesis, options are more expensive on low-price stocks; therefore, we expect a positive coefficient on this variable.

We present the estimation results in Table 4. We include skewness-related characteristics (*IVOL*, *ISKEW*, *IKURT*, and *MAX10*) and other option and stock characteristics as controls. The option characteristics include measures of option

Table 4
FM regressions of delta-hedged option returns

	Delta-hedged calls	Delta-hedged puts
<i>Intercept</i>	-0.593 (-2.15)	-0.202 (-0.91)
<i>log(stock price)</i>	0.346 (6.12)	0.203 (4.66)
<i>IVOL</i>	-1.885 (-0.89)	-1.134 (-0.73)
<i>ISKEW</i>	-0.001 (-0.23)	0.005 (1.13)
<i>IKURT</i>	-0.005 (-4.31)	-0.003 (-2.91)
<i>MAX10</i>	-1.292 (-1.06)	-1.910 (-1.89)
<i>Option volume/open interest</i>	-0.001 (-0.30)	0.004 (0.87)
<i>Option bid-ask spread</i>	-0.345 (-6.14)	-0.234 (-3.95)
<i>IV - HV</i>	-1.605 (-15.12)	-1.224 (-12.52)
<i>Gamma</i>	1.543 (4.73)	1.233 (3.95)
<i>Vega</i>	-0.031 (-4.10)	-0.010 (-2.27)
<i>log(size)</i>	-0.041 (-5.21)	-0.036 (-5.56)
<i>log(market-to-book)</i>	-0.043 (-3.46)	-0.038 (-2.93)
<i>Past stock return</i>	-0.044 (-1.15)	-0.032 (-1.06)
<i>Profitability</i>	0.037 (2.66)	0.039 (2.74)
<i>Forecast dispersion</i>	0.036 (2.14)	0.053 (2.97)
<i>Net equity issuance</i>	-0.096 (-1.62)	-0.034 (-0.63)
<i>Stock illiquidity</i>	-7.149 (-5.66)	-3.193 (-1.61)

Each month we run cross-sectional FM (1973) regressions of the delta-hedged weekly option returns. The main independent variable is the log of the underlying stock price. The control variables include option and stock characteristics as described in Table 1. All coefficients are multiplied by 100, and Newey and West (1987) corrected *t*-statistics (with twelve lags) are in parentheses. The sample period is 1996-2020.

liquidity such as the ratio of option volume to open interest and option bid-ask spread as well as option Greeks such as *Gamma* (the second derivative of the option price to the underlying price) and *Vega* (the sensitivity of the option price to stock return volatility) that can potentially reflect difference in the riskiness of options and, therefore, affect option returns. We also include *IV–HV* following Goyal and Saretto (2009), who document that this variable is a strong predictor in the cross-section of option returns. The stock characteristics include the *Logarithm of market capitalization*, the *Logarithm of the market-to-book ratio*, *Past 6 month stock return*, *Profitability*, *Analyst forecast dispersion*, *Net equity issuance*, and Amihud's (2002) illiquidity measure.

The regression coefficients reported in Table 4 on the various stock and option characteristics are generally consistent with the existing studies. Although the ratio of option volume to open interest does not affect subsequent option returns, the stock's Amihud illiquidity has a negative effect for both put and call returns, and the option *Bid-ask spread* has a negative effect for the delta-hedged portfolios. This is consistent with the idea that investors in options markets demand additional compensation for holding illiquid option positions if they have net short option positions. Note that Lakonishok et al. (2006) find that nonmarket maker investors in aggregate have more written than purchased options. Consistent with the findings of Goyal and Saretto (2009), *IV–HV* is negatively related to option returns for the delta-hedged call and put portfolios. Options' *Gamma* has a significant positive effect, while *Vega* has a significant negative effect, confirming the importance of controlling for specific aspects of the riskiness of option portfolios.

Size has a significant negative effect on portfolio returns, again indicating the importance of controlling for stock characteristics in our regressions (since size is clearly positively correlated with the stock price). The effect of the *Market-to-book ratio* is negative, although statistically insignificant. Amihud's (2002) measure of stock illiquidity is negative and highly significant in our regressions of option returns.⁸

The coefficient on the logarithm of the underlying *Stock price*, our main variable of interest, remains positive and highly statistically significant after including both stock and option characteristics as controls. The coefficient is 0.346 (*t*-statistic = 6.12) for calls and 0.203 (*t*-statistic = 4.66) for puts. This effect is also economically large. For example, the coefficient estimates imply that the returns to delta hedged call portfolios increase by 0.30% per week when the underlying stock price moves by one standard deviation.⁹

⁸ Vasquez and Xiao (2024) find that default risk is related to option returns. We control for the distress risk measure of Campbell, Hilscher, and Szilagyi (2008) and find that the coefficient on stock price remains similar to that in Table 4. Muravyev, Pearson, and Pollet (2022) show that stock borrow fees explain the predictability of stock returns from several option variables. Although our focus is on delta-hedged option returns, we nevertheless control for the spread in *IV* between calls and puts and find that the coefficient on stock price remains similar to that in Table 4. Finally, we also control separately for option open interest separately and/or stock bid-ask spread and find that the coefficient on stock price shows no material change.

⁹ We use options that are closest to ATM from a sample of options with moneyness between 0.8 and 1.2. As such, we permit the entry of OTM or in-the-money (ITM) options in some cases. The evidence on the volatility smile indicates

We provide many additional robustness tests in the [Internet Appendix](#). These include alternative measures of mispricing ([Table IA.1](#)), empirical choices for computing option returns (return to month-end, return to next week, and daily rebalancing ([Table IA.2](#)); various sub-sample splits ([Table IA.3](#)), trading OTM and ITM options, and options with time to maturity of more than 1 month ([Table IA.4](#))).

Role of Retail Investors

Our conjecture is that demand from retail investors pushes the prices of options on low-price stocks up and leads to option overpricing. We posit that retail investors have limited capital and hence are attracted to options on low-price stocks. In this section, we conduct two tests for the role of retail investors. First, we show that net buying of retail investors decreases as underlying stock-price increases, while the reverse is true for institutional investors. Second, we show directly that net retail demand is related to option expensiveness.

Retail traders' demand

From ISE, we obtain data that constitute the complete daily records of the option buy and sell activities of different market participants since 2009. We obtain these data from 2009 to 2017. [Ge, Lin, and Pearson \(2016\)](#) report that ISE trades represent about 30% of the total trading volume on individual equity options. Following the literature, we compute total net buy as the difference between the total daily buy and sell trade positions (measured by both the trading volume and the number of trades), divided by the total daily trade positions. Similarly, we compute the measures of net buy separately for retail and professional investors. Following the literature, we classify all types of customer traders as retail investors, and proprietary traders and brokers/dealers as professional investors. In our sample period, retail investors account for 84% (70%) of the number of trades (trading volume) in the ISE data. Market clearing implies that market makers take the opposite side, and the sum of retail and professional trades is equal to market maker volume. In addition to looking at all positions, we also examine only the opening positions, as these positions might be more indicative of investors' trading activity on a given date ([Pan and Potesman 2006](#)).

We acknowledge that the customer category in the ISE data contains some trades of professional traders (in addition to the trades of retail investors). In particular, some large customer trades may actually be executed on behalf of institutions. Unfortunately, the level of granularity of the ISE data does not allow for a more refined analysis. However, the professional category in the ISE data

that OTM and ITM options tend to be more expensive (from an implied volatility perspective) than ATM options. To check for this potentially confounding effect, we include moneyness or its absolute value as an additional control in FM regressions. We find that this does not materially change the coefficient of interest. For example, in regressions of delta-hedged calls, the coefficient on log stock price declines from 0.346 to 0.342 (0.283) with moneyness (absolute moneyness) as an additional control.

(proprietary traders and broker/dealers) is presumably not tainted by misclassification. Since, as we will shortly show, our results obtain for the customer category but not for the professional category, any contamination of institutional trades in the customer data would bias our results downward, working against our hypothesis.

We run cross-sectional FM regressions of these measures of net buy by all, retail (proxied by the customer category), or professional investors on the logarithm of the underlying stock price and a set of the same control variables as in Table 4. We argue that retail investors, constrained by their budgets, trade options on low-price stocks more actively than they do options on high-price stocks. On the other hand, preferences for skewness and divergence of opinion make retail investors net buyers of such options. Consequently, we expect an inverse relationship between the net buy of retail investors and the underlying stock price.

For the sake of brevity, we report the coefficients only on our main variable of interest, the *Logarithm of the stock price*. These coefficients are reported in Table 5. We find that all our measures of net buy (based on all positions, as well as opening positions, and using total trading volume, as well as the number of trades) for retail investors are negatively and highly statistically significantly associated with the underlying stock prices. We do not find the same effect for professional investors. The coefficients for opening positions are higher (in absolute value) than those for all positions. Therefore, there is also some evidence in our tests that opening positions are more informative than other positions.

Table 5
FM regressions of retail/professional trading on underlying stock price

	Call options			Put options		
	Total net buy	Retail net buy	Professional net buy	Total net buy	Retail net buy	Professional net buy
<u>Trading volume</u>						
All positions	-0.136 (-4.66)	-0.163 (-5.51)	0.140 (1.46)	-0.061 (-2.45)	-0.075 (-2.97)	1.655 (1.21)
Opening positions	-0.164 (-4.92)	-0.210 (-5.04)	-0.266 (-0.53)	-0.126 (-3.49)	-0.139 (-3.62)	0.867 (1.32)
<u>Quantity of trades</u>						
All positions	-0.130 (-4.30)	-0.158 (-5.05)	0.138 (1.39)	-0.061 (-2.40)	-0.072 (-2.70)	1.654 (1.20)
Opening positions	-0.141 (-4.06)	-0.188 (-4.57)	-0.263 (-0.53)	-0.119 (-2.55)	-0.125 (-2.60)	0.892 (1.33)

Each month we run cross-sectional FM (1973) regressions of net buy trades. Total net buy is the difference between the total daily buy and sell trade positions (measured by both trading volume and quantity of trades), divided by the total daily trade positions. Retail (professional) net buy is the difference between all retail (professional) daily buy and sell trade positions, divided by all retail (professional) daily trade positions. Retail investors include all types of customer traders, and professional investors include proprietary traders and brokers/dealers. The table also shows results for equivalent measures using position-opening trades only. The main independent variable is the *Log of the underlying stock price*. The control variables include the same set of option and stock characteristics as in Table 4. The table shows only the coefficients on the log(stock price), where Newey and West (1987) corrected *t*-statistics (with twelve lags) are in parentheses. The sample includes 46,137 call options and 36,056 put options over the period 2009-2017.

Option demand and option expensiveness

We have already shown that decile-1 options are more expensive than decile-10 options. Subsection Retail traders' demand shows that the options in decile 1 have excess retail demand. In this subsection, we tie these findings together by showing that it is the excess demand from retail investors that leads to option expensiveness. We follow [Gârleanu, Pedersen, and Poteshman \(2009\)](#) in measuring the net demand as net open interest. We consider stocks with strictly positive trading volume on at least 80% of days over the last year. For each of these stocks on each trading date we include all options (calls and puts) with 15-45 calendar days to expiration and moneyness between 0.95 and 1.05. For each of those options, we use the ISE database for classifications of open/close and buy/sell trading volume made by retail investors. We then measure the retail daily buy/sell open interest using the recursive equations starting from the first day of the option trade, following [Ni et al. \(2021\)](#):

$$\begin{aligned} OpenInterest_{j,t}^{Buy} &= OpenInterest_{j,t-1}^{Buy} + Volume_{j,t}^{Open\ Buy} - Volume_{j,t}^{Close\ Sell} \\ OpenInterest_{j,t}^{Sell} &= OpenInterest_{j,t-1}^{Sell} + Volume_{j,t}^{Open\ Sell} - Volume_{j,t}^{Close\ Buy}, \end{aligned} \quad (2)$$

for option j on trading day t . The retail investors' net demand on day t is then defined as the difference between the retail buy and sell open interest. We aggregate the net demand of all options for each trading day in a month to obtain the retail net demand at the monthly frequency for each firm. [Gârleanu, Pedersen, and Poteshman \(2009\)](#) consider weighting net demands based on different types of unhedgeable risks. We follow these authors and multiply the demand by options *Gamma* (to capture the impossibility of continuous-time trading) or *Vega* (to capture stochastic volatility risk) before aggregating to the monthly level. Finally, the aggregate monthly net demand is scaled by the firm's number of outstanding shares to make the measures comparable between firms. In summary, we consider three measures of net demand: *Raw*, *Gamma weighted*, and *Vega weighted*.¹⁰

Option expensiveness is measured as the difference between the option's implied volatility and the expected volatility derived from the GARCH(1,1) model based on the past 60 months (see [Gârleanu, Pedersen, and Poteshman 2009](#)). We average the option expensiveness for each trading day in the month, and then take another average across all trading days in the month to obtain the firm's option expensiveness at a monthly frequency.

We finally run cross-sectional FM regressions of *Option expensiveness* on *Net demand*. The control variables are also motivated by [Gârleanu, Pedersen, and Poteshman \(2009\)](#). These include the total trading volume of options on the firm's stock during the month, the return on the firm's stock in the past month and the past volatility of the stock return, measured by the standard deviation of the daily

¹⁰ We acknowledge potential contamination of the customer category with some professional trades. As discussed in Section , any such contamination would bias our results downward, working against our hypothesis.

Table 6
FM regressions of option expensiveness on net retail demand

	Raw net demand		Gamma-weighted net demand		Vega-weighted net demand	
	Decile 1	Decile 10	Decile 1	Decile 10	Decile 1	Decile 10
<i>Intercept</i>	-2.260 (-0.55)	2.980 (3.83)	-2.230 (-0.53)	2.610 (3.28)	-2.480 (-0.60)	2.860 (3.89)
<i>Net retail demand investors</i>	0.460 (7.34)	-0.030 (-1.18)	2.060 (5.10)	2.310 (3.04)	0.510 (5.35)	0.000 (0.56)
<i>Total option trading volume</i>	0.620 (2.24)	0.060 (1.29)	0.660 (2.41)	0.100 (1.82)	0.670 (2.33)	0.070 (1.60)
<i>Past stock return</i>	-11.570 (-1.12)	-7.230 (-2.72)	-11.570 (-1.10)	-7.180 (-2.77)	-11.940 (-1.22)	-7.390 (-2.80)
<i>Past stock return volatility</i>	-43.080 (-1.76)	-42.790 (-6.00)	-43.980 (-1.74)	-39.370 (-5.12)	-44.160 (-1.83)	-40.990 (-5.34)

The table shows FM (1973) firm-level monthly regressions of the option expensiveness on the net demand of retail investors for the options. *Option expensiveness* is measured as the difference between the option's implied volatility and the expected volatility derived from the GARCH(1,1) model. *Net demand* is measured as the difference between buy and sell open interest. We consider three versions of net demand: raw, Gamma weighted, and Vega weighted. All three versions are scaled by shares outstanding. The control variables include the total trading volume of the options on the firm's stock during the month, the return on the firm's stock in the past month, and the past volatility of the stock return, measured by the standard deviation of the daily returns over the past year. We run the regression separately for the top and bottom stock price deciles (decile 1 = lowest stock prices, decile 10 = highest stock prices). All coefficients are multiplied by 100, and Newey and West (1987) corrected *t*-statistics (with twelve lags) are in parentheses. The sample period is 2009-2017.

returns over the past year. We run these regressions separately for deciles 1 and 10 of the stock price. Given our earlier evidence in Table 5 of excess demand for options in decile 1, we expect to see a positive coefficient on net demand for decile 1 and a smaller (or 0) coefficient for decile 10.

Table 6 presents the results. We find that *Option expensiveness* is positively related to option volume. This is prima facie evidence in support of Garleanu, Pedersen, and Poteszman (2009). We also find that *Option expensiveness* is negatively related to *Past stock return* and *Stock volatility* for both deciles 1 and 10, although the coefficients are statistically significant only for decile 10. The coefficient of interest to us is the one on *Net retail investor demand*. We find that *Option expensiveness* is positively related to investor demand for decile 1. All coefficients are statistically significant regardless of the way we measure net demand. In contrast, the coefficient on *Net retail demand* is small and statistically insignificant for decile 10 for raw net demand and *Vega-weighted net demand*. The only exception is that *Option expensiveness* is positively related to *Gamma-weighted net demand* for decile 10.

Budget constraints: Quasi-natural experiments

In this section, we take advantage of three quasi-natural experiments to study the effect of underlying stock prices on option expensiveness. These experiments allow us to study the direct effect of underlying prices on option prices and returns;

that is, the effect that is not potentially confounded by additional firm characteristics. In addition, these experiments provide evidence in support of budget constraints of retail investors (Hypothesis 1). First, we take advantage of the mini option contracts. These are contracts for 10 shares of the underlying stock (as opposed to 100 shares in the case of standard option contracts). Mini option contracts are available for three stocks: Amazon, Apple, and Google. Second, we examine the effect of stock splits on option expensiveness. Third, we look at mini index options, available for Nasdaq 100 Mini-NDX and S&P500 Mini-SPX indices.

Mini options

In 2013, options exchanges launched mini options on three popular stocks: Amazon, Apple, and Google. As their name suggests, mini options are a smaller version of the standard option contract. Mini contracts carry a deliverable of 10 shares of an underlying security, unlike standard contracts of 100 shares. Strike prices were offered at the same price levels as standard contracts, and so were the minimum price variation for bids and asks. These mini options were later discontinued.

Mini options provide an ideal setting for studying the effect of budget constraints on option misvaluation. The underlying security is exactly the same for mini and standard options, and the only difference comes from the number of deliverable shares. Therefore, in the absence of budget constraints, the price of a mini option contract should be exactly one-tenth of the price of the corresponding standard contract. However, if retail investors are budget constrained, they will gravitate towards mini options, whose prices might then diverge from the prices of standard options on the same underlying. Our tests in this section are aimed at detecting differences between the prices of standard and mini options.

To implement these tests, each trading day we match a standard option contract with a mini option contract with the same expiration date and the strike price. For each pair of options, we compute the option *Price ratio* of the mini contract to that of the standard contract as:

$$\text{Price ratio} = \frac{\text{Mini option price} \times 10}{\text{Main option price}}. \quad (3)$$

As in our main tests, we include only options with moneyness (ratio of stock price to strike price) between 0.8 and 1.2. Table 7 shows that the prices of the mini option are significantly higher than those of the standard option for all three stocks. The price differences are also economically large and vary between 1.1% (calls on Google) to 5.4% (puts on Amazon). Since the underlying securities are identical for mini and standard option contracts, these results are strongly suggestive of the fact that price differences are due to the excess retail demand for mini options, driven by budget-constrained retail investors.

Table 7
Mini options

		Amazon	Apple	Google
Calls	# paired options	8,709	20,517	10,851
	Price ratio	1.038	1.013	1.011
	<i>t</i> -statistic ($H_0 = 1$)	(15.70)	(18.19)	(4.36)
Puts	# paired options	6,696	14,705	8,285
	Price ratio	1.054	1.023	1.024
	<i>t</i> -statistic ($H_0 = 1$)	(19.36)	(19.18)	(16.09)

The table shows comparisons of option prices when traded by standard contracts (100 shares per contract) and when traded by mini contracts (10 shares per contract) on the stocks of three major companies: Amazon, Apple, and Google. Each trading day we match an option traded by a standard contract with an option traded by a mini contract based on the expiration date and the strike price. For each pair of options, we compute the option price ratio of the mini contract to the standard contract, which equals:

$$Price\ ratio = \frac{Mini\ option\ price \times 10}{Main\ option\ price}.$$

We include only options with moneyness (ratio of stock price to strike price) between 0.8 and 1.2. All price ratios are winsorized at the top and the bottom percentiles. The *t*-statistics in parentheses are based on standard errors clustered by expiration day/strike price. The sample period is March 2013 to January 2016 for Amazon, March 2013 to January 2017 for Apple, and March 2013 to April 2014 for Google.

We note that the price differences between mini and regular options may appear striking because market makers typically manage inventory jointly for both types of options, which should prevent persistent violations of the law of one price through arbitrage. However, we view these price differences not as sustainable arbitrage opportunities but as evidence of retail trading pressure and associated limits to arbitrage. We formally investigate the role of limits to arbitrage in Section Limits to Arbitrage and Alternative Explanations.

Stock splits

Stock splits provide a natural laboratory to study the effect of the underlying stock price on the prices of options and their potential expensiveness. Many studies show that stock attractiveness increases following splits (see, for example, [Easley, O'Hara, and Saar 2001](#) and [Schultz 2000](#)). Stock splits may act as a signaling device by which management conveys information to outside investors about the fundamental prospects of the company. For example, [Lakonishok and Lev \(1987\)](#) find mean reversion in earnings growth around stock splits, although [Weld et al. \(2009\)](#) question the signaling aspect of splits. There is also some evidence of abnormal stock returns following splits (see, for example, [Desai and Jain 1997](#)). Thus, it may be that the postsplit stock price is not the same multiple of the presplit stock price as the split ratio.

We design a test statistic that is agnostic about the information content, if any, of stock splits. We posit that, absent budget constraints, the ratio of the price of an ATM option (either a put or a call) to the price of the stock should not change around the split. On the contrary, in the presence of budget constraints, postsplit options (on lower priced stocks) should be more expensive relative to presplit options (on higher priced stocks). This implies that the ratio of option to stock price

should increase following the split. Since we analyze the pre- and postsplit ratio of the option-to-stock price, our hypothesis is immune to the information effects on the underlying stock.

To test this hypothesis, we collect a sample of stock splits from CRSP. Our sample contains 1,981 stock splits (with split ratios of two or more) with traded options over the 1996-2020 sample period. For each stock split, we first identify the shortest maturity options among all options with at least 1 month to maturity, and then pick the option that is closest to the ATM. Panel A of Table 8 shows the summary statistics of the stock split sample. In most cases the split ratio was 2, though the distribution of the split ratio is highly skewed, with a maximum ratio of 50.¹¹ The average presplit stock price is \$97.99 which goes down to \$43.60 postsplit.

To formally test the hypothesis of investors' reaction to the underlying stock price, we examine the change to the option/stock-price ratio around the stock split. The main variable of interest is defined as:

$$\begin{aligned} & \text{Post/Pre-split option/stock ratio} \\ &= \frac{\text{Post-split option price/Pre-split option price}}{\text{Post-split stock price/Pre-split stock price}} \end{aligned} \quad (4)$$

for calls and puts separately. The null hypothesis is that this price ratio is equal to 1. If (retail) investors' budget constraints result in relative expensiveness of option on low-price stocks, then the price ratio should be greater than 1 as option prices react to the split and the consequent change in the underlying stock prices. We analyze windows of 1, 3, and 5 days before and after the stock split day.

Stock return volatility typically increases after splits (see, for example, Brandt et al. 2010 and Ohlson and Penman 1985). However, stock splits in our sample are anticipated events as the date of the split is announced well in advance. Therefore, presplit implied volatility, which is a forward-looking measure of volatility (absent volatility risk premium effects), should already reflect the expectation of increased volatility. Even if investors are unaware of the fact that volatility increases after split, our use of short windows around the event means that investors use the same expected volatility pre- and postevent (higher volatility if they are aware of the empirical patterns in volatility; lower volatility otherwise). In other words, over the short windows that we analyze, we do not expect changes in the implied volatility of options.

Panel B of Table 8 presents the results from the *t*-test for the null hypothesis that the change in the *Option-to-stock-price ratio* is equal to 1. We find that the change in the *Option-to-stock-price ratio* is always greater than 1, across all time windows and for both puts and calls. For example, call options become 13% more expensive relative to the underlying price within a 5-day window around a stock split, while put options become 16% more expensive over the same window.

¹¹ When we restrict the sample to include only 2-for-1 splits, there is almost no change in our results.

Table 8
Changes in option prices around stock splits

A. Summary statistics

	Mean	Std	Min	P25	P50	P75	Max
<i>Split ratio</i>	2.23	1.28	2.00	2.00	2.00	2.00	50.00
<i>Presplit stock price</i>	97.99	115.43	12.06	60.00	80.00	110.59	3476.00
<i>Postsplit stock price</i>	43.60	25.73	5.07	28.74	38.09	51.28	498.32

Window around stock split

(-1, +1) (-3, +3) (-5, +5)

B. Post/presplit option/stock price ratio

Calls	Mean ratio	1.07	1.13	1.13
	<i>t</i> -statistic ($H_0 = 1$)	(7.80)	(7.47)	(5.61)
Puts	Mean ratio	1.07	1.09	1.16
	<i>t</i> -statistic ($H_0 = 1$)	(5.55)	(3.68)	(4.30)

C. Post/presplit standardized IV ratio

Calls	Mean ratio	1.13	1.12	1.05
	<i>t</i> -statistic ($H_0 = 1$)	(3.99)	(3.32)	(4.85)
Puts	Mean ratio	1.09	1.08	1.07
	<i>t</i> -statistic ($H_0 = 1$)	(5.45)	(3.84)	(5.43)

D. Mean abnormal returns (in %)

Calls	0.09	0.21	0.23
	(1.77)	(1.93)	(1.20)
Puts	0.29	0.61	1.25
	(6.01)	(5.26)	(5.79)

E. Regression of option returns

Calls	Intercept	-0.46	-1.54	-2.26
		(-2.51)	(-4.04)	(-3.70)
	log(split ratio)	0.73	2.29	3.24
Puts	Intercept	-0.15	-0.61	-1.62
		(-0.88)	(-1.50)	(-2.26)
	log(split ratio)	0.56	1.59	3.76
		(2.71)	(3.12)	(4.16)

The sample contains 1,981 stock splits over the period 1996-2020 with options traded on the stock. For each stock split, we pick one option; we first identify the shortest maturity options among all options with at least 1 month to maturity, and then pick the option that is closest to ATM. Panel A shows statistics of the stock split sample. In panel B we present the change to the option/stock price ratio around the stock split, calculated as:

$$Post/Pre-split option/stock price ratio = \frac{Post-split option price / Pre-split option price}{Post-split stock price / Pre-split stock price}.$$

We look at a window of 1-5 days before and after the stock split day. Panel C presents the change to the standardized implied volatility of a firm's options around the stock split. Panel D shows the mean option abnormal return around the stock split, where abnormal return is the return on the option's delta-hedged position minus the return on the equivalent position on the S&P 500 index. Panel E shows regressions of the option *Abnormal return* on the (log of) the *Split ratio*. All price ratios and abnormal returns are winsorized at the top and bottom percentiles.

If the split has (on average) an effect on the valuation of the stock (Desai and Jain 1997), then this will change the moneyness of the option in a systematic way. Since OTM and ITM options tend to be more expensive than ATM options (the

volatility smile), this could potentially impact the results in panel B of Table 8. Keeping maturity and moneyness constant alleviates this concern. Accordingly, we also use the implied volatility of the stock taken from OptionMetrics' implied volatility surface. Panel C of Table 8 reports these post/presplit standardized IV ratios. We find again that both calls and puts are more expensive (as evidenced by higher IVs) postsplit.¹² We reiterate that our evidence of change in option prices (implied volatility) cannot be attributed to changes in realized volatility as documented in the prior literature but, rather, is more likely due to budget constraints of retail investors.

We also analyze abnormal option returns around stock splits. If options become more expensive due to investors' demand for options with low underlying prices, then both calls and puts should exhibit positive abnormal returns around a stock split. We measure abnormal option returns as the return on the option's delta-hedged position minus the return on the equivalent position on the S&P 500 index. We calculate these excess returns over windows of 1, 3, and 5 days around the split. Panel D of Table 8 presents the results of this test. As expected, all option abnormal returns are positive. The results are stronger for puts than for calls. The abnormal returns to delta-hedged puts are highly statistically significant for all three return windows. For example, abnormal returns for delta-hedged puts are 1.25% over a 5-day window around the stock split. For calls, the results are weaker, with abnormal returns being marginally statistically significant.

We also expect that the magnitude of abnormal returns should depend on the split ratio—the higher the split ratio, the greater the potential increase in option expensiveness postsplit. To formally test this prediction, we regress abnormal option returns on the *Logarithm of the split ratio* and present the results in panel E of Table 8. We find that the coefficients on the *Split ratio* are positive and highly significant for all windows.

Overall, evidence from stock splits again points towards the role of budget constraints in shaping option expensiveness: following a stock split, budget-constrained retail investors buy more options on the stock, thereby increasing their prices.

Mini index options

To further document the role of budget constraints, we take advantage of another feature of the market: options written on mini indices. Mini index options are a scaled down version of main index options—they are index options with only one-tenth the contract size of main index options. The lower cost of mini index options

¹² Shue and Townsend (2021) also document an increase in implied volatility around splits. Their focus is on longer windows around the event. They attribute the increase in implied volatility to option traders not fully incorporating expected increase in realized volatility into their forecasts. Shue and Townsend also show that delta-hedged straddle positions are profitable in the 40 days following the stock split. As mentioned earlier, our use of a short window around the split dates allows us to sidestep the issue of investors updating their volatility expectations in the long term. See also Dubinsky et al. (2018) who show that anticipated uncertainty in earnings commands a premium in the options markets.

gives small retail option traders the opportunity to become more involved in trading on the broader market.

Mini index options were first introduced to the market in the form of Mini-NDX Index Options in the year 2000. Mini-NDX Index Options are options offered on a one-tenth scaled down index of the NDX (Nasdaq100) known as the MNX. Following the introduction of Mini-NDX, Mini-SPX Index Options were offered in 2005 on a one-tenth scaled down index of the SPX (S&P 500) known as the XSP. Options on both the main and the mini index are European, and the only difference between the two types of options is the scale-down factor.¹³

Note that mini index options offer an even cleaner setting than stock splits as there are no countervailing effects of signaling, changing fundamentals, or changing volatility of stocks postsplit. Absent any excess demand for mini index options, the prices of mini index options should be one-tenth of the prices of the main index options as the only difference between the main index and the mini index is the scaling factor.

To analyze the difference between main and mini index options, on each trading day, we match an option on the main index with an option on the mini index based on the expiration date and the adjusted strike price. We approximate prices with midpoint bid-ask quotes. For each pair of options, we compute the *price ratio* as:

$$\text{Price ratio} = \frac{\text{Mini-index option price} \times 10}{\text{Main-index option price}}. \quad (5)$$

We include only options with moneyness (ratio of index to strike price) between 0.9 and 1.1. We present the results for different days to maturity in Table 9. We have about 70,000 observations for both call and put options.

We find that, for both put and call options, the scaled ratio of mini index option price to main index option price is greater than 1 (and is highly statistically significantly different from 1), indicating relative expensiveness of mini index options. We also find more mispricing for puts than for calls. For example, the average overpricing is approximately 0.9% for call options and about 1.3% for put options with less than 30 days to maturity. The mispricing is more pronounced for short-term options than for longer dated options. Mispricing declines from 1.3% to 0.1% for put options with less than 30 days to maturity to those with 91 to 180 days to maturity.

In summary, looking at mini option contracts, stock splits, and mini index options reveals similar patterns in the relation between option prices and the cost of buying the options, whereby options on securities with lower prices as well as mini contract options tend to be relatively more expensive, supporting the role of budget constraints.

¹³ Note that mini index options are different than the mini contract options analyzed in Section 4.1; for mini index options, the underlying asset is different (main index vs. mini index), whereas for mini contract options, the underlying asset is the same; only the size of the contract is smaller.

Table 9
Mini index options

		Days to maturity		
		Up to 30	31 to 90	91 to 180
Calls	# paired options	33,042	33,292	3,967
	Price ratio	1.009	1.002	1.002
	<i>t</i> -statistic ($H_0 = 1$)	(11.73)	(4.99)	(3.20)
Puts	# paired options	37,837	37,443	5,162
	Price ratio	1.013	1.003	1.001
	<i>t</i> -statistic ($H_0 = 1$)	(12.39)	(11.83)	(2.70)

The table shows comparisons of option prices on major stock price indices and their traded mini indices, specifically, the Nasdaq 100 Mini-NDX Index Options (MNX) and the S&P 500 Mini-SPX Index Options (XSP). Both mini indices are one-tenth of the major indices. Each trading day we match an option on the main index with an option on the mini index based on the expiration date and the adjusted strike price. For each pair of options, we compute the *Price ratio*, which equals:

$$\text{Price ratio} = \frac{\text{Mini-index option price} \times 10}{\text{Main index option price}}.$$

We include only options with moneyness (ratio of index to strike price) between 0.9 and 1.1 and present the results for different days to maturity. All price ratios are winsorized at the top and the bottom percentiles. The *t*-statistics in parentheses are based on standard errors clustered by expiration day/strike price. The sample period for the Nasdaq 100 options is 2000-2020, and for the S&P 500 options is 2013-2020.

Exploring Possible Explanations

We argue in Section Hypotheses Development that while budget constraints are likely to motivate retail investors to trade options on low-price stocks, some additional mechanism must create positive net demand for such options from retail investors. In that section, we propose two potential mechanisms: preference for skewness (Hypothesis 2a) and divergence of opinions (Hypothesis 2 b). In this section, we provide tests for both of these hypotheses.

Skewness preference

The literature documents that investors have preference for low-price assets due to skewness preference (Barberis and Huang 2008; Mitton and Vorkink 2007) or lottery preferences (Bali, Cakici, and Whitelaw 2011; Kumar 2009). Preference for lottery-like stocks might also translate into preference for options on such stocks (Blau, Bowles, and Whitby 2016; Byun and Kim 2016). We posit that retail investors with limited budgets trade options on low-price stocks. If such investors have preferences for positively skewed payoffs, they will then likely buy call options on such stocks, thereby increasing their prices. Relative expensiveness of call options will then lead to relative expensiveness of put options via the put-call parity. Furthermore, skewness-seeking retail investors will be more likely to buy call options on stocks with high underlying returns skewness as such contracts offer even more positively skewed returns (Hypothesis 2a).

To test for the effect of skewness preferences, each month, we sort all options independently into quintiles based on the market price of the underlying stock and

Table 10
Effect of skewness

Price ↓	Low	Q2	Q3	Q4	High	High – Low
<i>A. ISKEW</i>						
Calls						
Low	– 0.47 (– 9.21)	– 0.35 (– 6.14)	– 0.40 (– 7.55)	– 0.46 (– 9.10)	– 0.57 (– 10.35)	– 0.10 (– 2.07)
High	– 0.06 (– 2.44)	– 0.07 (– 2.58)	– 0.08 (– 3.08)	– 0.09 (– 2.98)	– 0.10 (– 3.21)	– 0.03 (– 1.36)
Puts						
Low	– 0.24 (– 5.86)	– 0.25 (– 5.72)	– 0.18 (– 4.07)	– 0.31 (– 7.29)	– 0.31 (– 7.45)	– 0.07 (– 1.76)
High	– 0.01 (– 0.45)	– 0.04 (– 1.62)	– 0.07 (– 2.61)	– 0.05 (– 1.98)	– 0.06 (– 1.80)	– 0.05 (– 1.72)
<i>B. MAX10</i>						
Calls						
Low	– 0.21 (– 2.83)	– 0.22 (– 3.08)	– 0.25 (– 3.92)	– 0.40 (– 4.88)	– 0.63 (– 6.91)	– 0.43 (– 4.11)
High	– 0.07 (– 2.25)	– 0.07 (– 1.89)	– 0.08 (– 1.77)	– 0.12 (– 1.73)	– 0.11 (– 1.30)	– 0.04 (– 0.51)
Puts						
Low	– 0.07 (– 0.98)	– 0.12 (– 1.91)	– 0.13 (– 2.51)	– 0.15 (– 2.46)	– 0.37 (– 5.25)	– 0.31 (– 3.78)
High	– 0.03 (– 1.07)	– 0.02 (– 0.45)	0.01 (0.28)	– 0.01 (– 0.22)	– 0.09 (– 1.12)	– 0.06 (– 0.74)

Each month, we sort all options independently into quintiles based on the market price of the underlying stock and based on a stock characteristic, yielding 25 stock price/characteristic portfolios. Delta-hedged option returns are calculated as in Table 2. The table reports the 7-factor alphas of selected portfolios. The factors are the Fama and French (2015) five factors, the Carhart (1997) momentum factor, and the Coval and Shumway (2001) zero-beta S&P 500 straddle factor. The rows entitled “Low” and “High” correspond to the extreme quintiles of stock prices. The columns correspond to the second stock characteristic, which is ISKEW in panel A and MAX10 in panel B. Variables are defined in Table 1. The table also shows the difference between the top and bottom quintiles for each characteristic. All alphas are in percent per week, and the corresponding *t*-statistics are in parentheses. The sample period is 1996-2020.

based on two proxies for the underlying return skewness, *ISKEW* and *MAX10*, yielding 25 stock-price/underlying skewness portfolios. We then report the weekly 7-factor alphas of these portfolios. Table 10 shows the results. Panel A uses *ISKEW* as a proxy for the underlying return skewness, while panel B is based on *MAX10*. To conserve space, we report results only for the lowest and highest quintiles of stock prices in the two rows marked Low and High, respectively.

Table 10 shows that delta-hedged options written on low-price stocks generate abnormally low returns regardless of the underlying skewness. Indeed, all option returns in the lowest quintile of the underlying stock price are negative and nearly all are highly statistically significant. In contrast, the returns to delta-hedged portfolios in the highest price quintile are always much smaller in magnitude and statistically insignificant for some portfolios.

More importantly, the underlying return skewness appears to affect the option returns in the lowest stock-price quintile in a direction consistent with Hypothesis 2a. Within that quintile, options in the top underlying skewness quintile underperform those in the bottom skewness quintile. The differences are larger for *MAX10* as a measure of skewness in panel B than those for *ISKEW* in panel A. For example, the return spreads between options in top and bottom skewness quintiles in panel B are -0.43% and -0.31% per week for call and put options, respectively. The corresponding spreads in panel A for *ISKEW* are -0.10% and -0.07% (the latter is only marginally statistically significant).

Note that there is virtually no effect of the underlying skewness on option returns in the top stock-price quintile. For example, the return spread between the top and the bottom *ISKEW* quintile is only -0.04% and -0.06% for calls and puts, respectively, for the quintile of the highest priced stocks. Since retail investors are less active in options on high-price stocks (as shown in Section Retail traders' demand), and institutional investors are less likely to have preferences for skewness and lottery-like payoffs, underlying skewness is unlikely to have a material effect on option returns for options on high-price stocks. This is exactly what we find.

Divergence of opinion

Alternatively, as shown by Miller (1977),¹⁴ divergence of opinion coupled with short sale constraints can potentially generate overpricing of securities. As discussed in Section Hypotheses Development, we posit that retail investors wishing to short options face multiple hurdles: they must obtain proper trading permissions and post substantial amounts in their accounts to comply with margin requirements. Naturally, it is much easier and less capital intensive for retail investors to buy options. This asymmetry in retail demand together with divergence of opinion can potentially create overpricing of options on low-price stocks (Hypothesis 2 b). We expect a greater degree of mispricing when the quality of information about the firm is low and the investors' opinions are likely more divergent. Because retail investors are most active in options on low-price stocks due to budget constraints, the effect should be contained to those options.

To test for the effect of divergence of opinion, each month, we follow the same procedure as in Subsection Skewness preference by sorting all options independently into quintiles based on the market price of the underlying stock and based on proxies for information transparency. We use four proxies for the degree of information uncertainty/divergence of opinion. First, we rely on the role of institutional ownership. Institutions facilitate information production, so higher institutional ownership is likely associated with less information uncertainty. Furthermore, as demonstrated by Boehmer and Kelley (2009), stocks with greater institutional ownership are priced more efficiently. Our second proxy for

¹⁴ For additional theoretical developments see also Duffie, Gârleanu, and Pedersen (2002) and Scheinkman and Xiong (2003).

information uncertainty is idiosyncratic stock return volatility (*IVOL*). Company valuations are likely less precise when stock return volatility is high. Next, we use the number of analysts who follow the stock and analyst forecast dispersion as proxies for information uncertainty. Firms followed by fewer analysts and also by analysts who disagree in their forecasts are likely to be less transparent. These information uncertainty proxies have been used extensively in the literature. For example, [Barron and Stuerke \(1998\)](#) argue that forecast dispersion is a useful indicator of uncertainty about the price-relevant component of firms' future earnings.

[Table 11](#) shows the results. For all four measures of divergence of opinion, in the quintile of options on low-price stocks, there is a negative effect of divergence of opinion on delta-hedged options returns. The spreads between the 7-factor alphas of the delta-hedged returns of the high- and low-information uncertainty portfolios are large in magnitude and highly statistically significant. For example, for institutional ownership as a proxy for information uncertainty, the spread is 0.31% and 0.30% per week for calls and puts, respectively. The corresponding spreads when using *IVOL* as a measure of information uncertainty are 0.44% and 0.33% for calls and puts, respectively.

Remarkably, and consistent with our hypothesis about the role of retail investors, there is no evidence that information uncertainty negatively affects option returns in the quintile of options on high-price stocks. In that quintile, most of the return spreads are insignificant, and some even have the opposite sign.

In general, the results in [Table 11](#) strongly support Hypothesis 2 b. There is an economically large and statistically significant effect of information uncertainty on option returns, which is limited to the sample of options on low-price stocks.

Limits to Arbitrage and Alternative Explanations

Our analysis thus far provides compelling evidence that options on low-price stocks appear relatively expensive, with this effect likely driven by retail investors' preference for skewness and divergence of opinion. However, two important questions arise. First, if these options are truly overpriced, why do sophisticated traders not eliminate this mispricing through arbitrage? Second, could our results be an artifact of the way option market makers set their quotes, rather than reflective of genuine expensiveness? In this section, we explore these alternative explanations and examine their implications.

Transaction costs

Our main results are based on the calculation of option returns from midpoint bid-ask quotes. The actual trading prices might be different and, therefore, influence our results. The prior literature shows that transaction costs in option markets are high (see, for example, [Figlewski 1989](#); [George and Longstaff 1993](#); and [Ofek, Richardson, and Whitelaw 2004](#)). We consider various scenarios of trading at prices that differ from midpoint quotes. In particular, we consider trading at

Table 11
Effect of dispersion of opinion

Price ↓	Low	Q2	Q3	Q4	High	High – Low
<i>A. Institutional ownership</i>						
Calls						
Low	-0.58 (-11.11)	-0.33 (-6.17)	-0.25 (-4.51)	-0.26 (-4.88)	-0.27 (-3.92)	0.31 (5.16)
High	-0.02 (-0.60)	-0.04 (-1.44)	-0.06 (-2.14)	-0.08 (-2.78)	-0.12 (-3.84)	-0.10 (-2.64)
Puts						
Low	-0.40 (-9.56)	-0.17 (-3.82)	-0.11 (-2.52)	-0.13 (-2.79)	-0.10 (-1.90)	0.30 (6.38)
High	-0.01 (-0.32)	-0.04 (-1.23)	-0.02 (-0.75)	-0.06 (-2.12)	-0.07 (-2.24)	-0.06 (-1.65)
<i>B. IVOL</i>						
Calls						
Low	-0.15 (-2.08)	-0.24 (-3.35)	-0.24 (-3.39)	-0.39 (-6.25)	-0.59 (-7.48)	-0.44 (-6.04)
High	-0.08 (-2.53)	-0.09 (-2.42)	-0.08 (-1.63)	-0.10 (-1.60)	-0.13 (-1.56)	-0.06 (-0.69)
Puts						
Low	-0.08 (-0.99)	-0.01 (-0.18)	-0.12 (-1.96)	-0.20 (-3.31)	-0.41 (-5.89)	-0.33 (-3.90)
High	-0.04 (-1.29)	-0.04 (-0.95)	0.02 (0.46)	0.00 (-0.03)	-0.04 (-0.56)	0.00 (-0.04)
<i>C. Analyst forecast dispersion</i>						
Calls						
Low	-0.61 (-9.70)	-0.25 (-3.19)	-0.31 (-4.72)	-0.29 (-5.69)	-0.37 (-6.77)	0.25 (5.56)
High	-0.13 (-3.42)	-0.12 (-4.39)	-0.06 (-1.90)	-0.06 (-1.48)	-0.07 (-1.37)	0.06 (1.44)
Puts						
Low	-0.40 (-7.98)	-0.14 (-2.00)	-0.11 (-1.80)	-0.08 (-1.66)	-0.17 (-3.59)	0.23 (5.58)
High	-0.08 (-2.17)	-0.08 (-2.48)	-0.02 (-0.52)	-0.02 (-0.49)	-0.05 (-0.97)	0.03 (0.70)
<i>D. Number of analysts</i>						
Calls						
Low	-0.54 (-10.20)	-0.43 (-8.27)	-0.29 (-4.93)	-0.10 (-1.46)	-0.18 (-2.34)	0.36 (4.72)
High	0.01 (0.17)	-0.03 (-0.86)	-0.09 (-2.83)	-0.07 (-2.54)	-0.08 (-2.88)	-0.09 (-2.03)
Puts						
Low	-0.39 (-9.02)	-0.27 (-6.48)	-0.14 (-2.75)	-0.05 (-1.06)	-0.05 (-0.90)	0.33 (5.96)

(continued)

Table 11
Continued

Price ↓	Low	Q2	Q3	Q4	High	High – Low
High	– 0.04 (– 0.82)	– 0.03 (– 0.88)	– 0.06 (– 2.17)	– 0.06 (– 2.13)	– 0.05 (– 1.84)	– 0.01 (– 0.26)

Each month, we sort all options independently into quintiles based on the market price of the underlying stock and based on a stock characteristic, yielding 25 stock price/characteristic portfolios. Delta-hedged option returns are calculated as in Table 2. The table reports the 7-factor alphas of selected portfolios. The factors are the Fama and French (2015) five factors, the Carhart (1997) momentum factor, and the Coval and Shumway (2001) zero-beta S&P 500 straddle factor. The rows entitled “Low” and “High” correspond to the extreme quintiles of stock prices. The columns correspond to the second stock characteristic, which is *Institutional ownership* in panel A, *IVOL* in panel B, *Analyst forecast dispersion* in panel C, and *Number of analysts* in panel D. *Institutional ownership* is measured by the sum of all shares held by institutions divided by total shares outstanding. Other variables are defined in Table 1. The table also shows the difference between the top and bottom quintiles for each characteristic. All alphas are in percent per week, and the corresponding *t*-statistics are in parentheses. The sample period is 1996-2020.

various values of the effective bid-ask spread (*ESPR*) measured in percentage of the quoted bid-ask spread (*QSPR*). For example, if the bid price of an option is \$3 and the ask price is \$4, then, assuming $ESPR/QSPR = 50\%$, we buy the option at \$3.75 (for the long leg of the 10–1 portfolio) and sell the option at \$3.25 (for the short leg of the 10–1 portfolio). The effective bid-ask spread of 0 then corresponds to our main results reported in Table 2. Recall that we keep the options until expiration; therefore, these adjustments apply only to prices at the initiation of positions.

The descriptive statistics in Table 1 show that the mean bid-ask spread is higher for the options in decile 1 than it is for those in decile 10. Therefore, we expect the transaction costs to have a substantial impact on the options in decile 1, but an inconsequential impact on the options in decile 10. We report both the mean excess returns and the 7-factor alphas from these tests in Table 12. These returns and alphas are reported for the top decile 10 (high stock price), the bottom decile 1 (low stock price), and for the 10–1 portfolio.

As expected, the returns and alphas to the short (decile 1) portfolio increase, and become less negative, as *ESPR* increases (when we sell at a lower price), while returns to the long (decile 10) portfolio decrease. The effects are more pronounced for decile one than for decile 10. For example, the return to delta-hedged calls decreases (in absolute value) from –0.75% for 0 spread to only –0.37% for $ESPR/QSPR = 50\%$. Similarly, the return to delta-hedged puts decreases (in absolute value) from –0.43% for 0 spread to only –0.17% for $ESPR/QSPR = 50\%$. The mean excess returns of the 10–1 portfolio remain highly statistically significant for an *ESPR* less than 25% of a *QSPR* for both delta-hedged calls and puts. However, for trades at prices beyond this range, for example at $ESPR/QSPR = 50\%$, the transaction costs prevent the profitable execution of our strategy for puts as the mean returns of 10–1 puts portfolios are only 0.07% (*t*-statistic = 1.62) for puts. At these high levels of transaction costs, call 10–1 portfolio average returns, while small at 0.21%, remain statistically significantly different from 0.

Table 12
Effect of transaction costs

	Delta-hedged calls			Delta-hedged puts		
	1	10	10 – 1	1	10	10 – 1
<i>A. Average return</i>						
<i>Mid bid-ask spread</i>	–0.75 (–10.78)	–0.12 (–2.99)	0.63 (11.02)	–0.43 (–8.43)	–0.07 (–1.87)	0.36 (8.84)
<i>ESPR/QSPR = 10%</i>	–0.67 (–9.74)	–0.13 (–3.15)	0.54 (9.69)	–0.38 (–7.38)	–0.08 (–2.01)	0.30 (7.45)
<i>ESPR/QSPR = 25%</i>	–0.55 (–8.15)	–0.14 (–3.42)	0.41 (7.60)	–0.30 (–5.84)	–0.09 (–2.27)	0.22 (5.31)
<i>ESPR/QSPR = 50%</i>	–0.37 (–5.47)	–0.16 (–3.89)	0.21 (3.93)	–0.17 (–3.22)	–0.10 (–2.70)	0.07 (1.62)
<i>B. 7-factor alpha</i>						
<i>Mid bid-ask spread</i>	–0.67 (–12.59)	–0.09 (–3.39)	0.58 (10.69)	–0.39 (–9.95)	–0.05 (–1.93)	0.34 (8.36)
<i>ESPR/QSPR = 10%</i>	–0.59 (–11.31)	–0.10 (–3.59)	0.49 (9.30)	–0.34 (–8.61)	–0.05 (–2.13)	0.28 (7.00)
<i>ESPR/QSPR = 25%</i>	–0.48 (–9.29)	–0.11 (–4.00)	0.37 (7.13)	–0.26 (–6.62)	–0.06 (–2.52)	0.20 (4.88)
<i>ESPR/QSPR = 50%</i>	–0.30 (–5.84)	–0.13 (–4.67)	0.17 (3.37)	–0.13 (–3.22)	–0.08 (–3.16)	0.05 (1.24)

We replicate the portfolio sort of [Table 2](#) while assuming different trading prices along the option's *Bid-ask spread*. For the row “Mid bid-ask spread,” we assume that the options are transacted at the midpoint of the bid and ask quotes; that is, the effective spread is 0 (the results from [Table 2](#)). The other rows correspond to different assumptions on the ratio of the effective bid-ask spread (*ESPR*) to the quoted bid-ask spread (*QSPR*). That is, we assume that we buy the options in the top decile (decile 10 = high stock prices) at a price higher than the mid bid-ask spread and sell the options in the bottom decile (decile 1 = low stock price) at a price lower than the *Mid bid-ask spread*. The table reports the *Mean excess return* and *7-factor alpha* for the top and bottom deciles and the long/short hedge portfolio. The factors are the [Fama and French \(2015\)](#) five factors, the [Carhart \(1997\)](#) momentum factor, and the [Coval and Shumway \(2001\)](#) zero-beta S&P 500 straddle factor. All alphas are in percent per week, and the corresponding *t*-statistics are in parentheses. The sample period is 1996–2020.

The effect of transaction costs is similar for the 7-factor alphas. The alphas are positive and statistically significant for an *ESPR* up to 25% of a *QSPR* for both delta-hedged calls and puts, and for delta-hedged calls, for *ESPR/QSPR* = 50%. However, the alphas are economically small (and statistically insignificantly different from 0) for puts for *ESPR/QSPR* = 50% (they remain statistically significant for *ESPR/QSPR* = 40%; results not reported).

Note that in actual practice, option traders would have the entire day to decide when to optimally trade and minimize the market impact costs. For example, [Muravyev and Pearson \(2020\)](#) argue that once trade timing is taken into account, effective spreads are much lower than the usually measured spreads. They estimate that effective spreads are only 30% of the quoted spread for such timers, albeit for a sample of S&P 500 stocks. At these spread levels, the posttransaction cost returns and the 7-factor alphas of the 10–1 portfolio are significant for delta-hedged calls and puts.

We also take the liberty to highlight that our main results ignoring transaction costs are still informative about behavioral biases. As argued by

Detzel, Novy-Marx, and Velikov (2023), while transaction costs do prevent arbitrageurs from exploiting an anomaly, it is only the analysis of gross returns (and not net returns) that “can provide economically useful information about how assets are priced in the market” (p. 1746).

Liquidity tests

While our results suggest that options on low-price stocks are relatively expensive, it is important to investigate whether this effect is primarily driven by illiquid stocks and options, where limits to arbitrage are typically higher. Liquidity is particularly relevant in the options market, as market makers might not quote competitive spreads for illiquid contracts. Although bid prices cannot go below 0, ask prices are unbounded, potentially leading to upward bias in midpoint prices and consequently more negative option returns. (Illiquidity can also give rise to pick-off risk, which we discuss in the next subsection.)

To address these concerns, we conduct several tests. First, we exclude options with bid prices below \$0.40, which helps filter out highly illiquid options that are subject to potential bias in option fair values relative to midpoint bid-ask quotes, as discussed above. Second, we separately analyze options on S&P 500 stocks and stocks ranked between 1001–2000 by market capitalization. Options on S&P 500 stocks are typically more actively traded and have tighter bid-ask spreads than those on smaller stocks, making them significantly more liquid than options on stocks ranked 1001–2000. Although our simple market cap ranking differs somewhat from the methodology used by Russell to construct the Russell 2000 index, the difference is minimal for our purposes, since these additional criteria primarily affect very small stocks that fall well below our 1001–2000 rank range.

The results, presented in Tables 13 and 14, show that filtering out low-bid options only marginally weakens our main findings. The 7-factor alpha for delta-hedged calls decreases from 0.58% to 0.38% per week but remains highly statistically significant (t -statistic = 7.82). For delta-hedged puts, the alpha similarly decreases from 0.34% to 0.27% per week while maintaining statistical significance.

Interestingly, the effects largely disappear for S&P 500 stocks, with the 7-factor alpha becoming statistically insignificant for both delta-hedged puts and calls. In contrast, the results become substantially stronger for stocks ranked 1001–2000 by market capitalization, with the 7-factor alpha increasing to 0.95% per week for delta-hedged calls and 0.52% per week for delta-hedged puts. These results suggest that market liquidity and associated limits to arbitrage play an important role in the formation of our results. However, it is important to note that S&P 500 stocks and their options tend to have higher institutional participation, with retail trading representing a smaller fraction of total volume compared to smaller stocks. Given that our hypothesis centers on the demand of retail investors, this difference

Table 13
Delta-hedged option portfolio returns for options with bid quotes above \$0.4

	1	2	3	4	5	6	7	8	9	10	10 – 1	10 – 1 opt-vw
A. Delta-hedged calls												
Mean excess return	-0.54 (-8.19)	-0.28 (-5.00)	-0.21 (-3.80)	-0.10 (-1.58)	-0.10 (-1.60)	-0.14 (-3.13)	-0.13 (-2.44)	-0.10 (-1.92)	-0.13 (-3.16)	-0.13 (-3.04)	0.41 (8.18)	0.36 (4.49)
7-factor alpha	-0.47 (-9.53)	-0.21 (-5.82)	-0.14 (-3.99)	-0.07 (-1.57)	-0.06 (-1.24)	-0.09 (-3.48)	-0.06 (-1.77)	-0.06 (-1.35)	-0.09 (-3.96)	-0.09 (-3.39)	0.38 (7.82)	0.34 (4.19)
B. Delta-hedged puts												
Mean excess return	-0.35 (-7.35)	-0.21 (-4.40)	-0.13 (-2.84)	-0.06 (-0.98)	-0.06 (-1.15)	-0.09 (-2.27)	-0.11 (-2.61)	-0.11 (-2.68)	-0.10 (-2.80)	-0.07 (-1.77)	0.29 (7.28)	0.28 (4.02)
7-factor alpha	-0.31 (-8.41)	-0.15 (-4.95)	-0.07 (-2.56)	-0.05 (-1.07)	-0.04 (-0.91)	-0.05 (-2.17)	-0.07 (-2.92)	-0.07 (-3.30)	-0.08 (-3.62)	-0.04 (-1.68)	0.27 (6.79)	0.24 (3.24)

Portfolios are sorted as in Table 1 (decile 1 = lowest stock price, decile 10 = highest stock price) except that we discard options with bid prices below \$0.4. For each option we calculate the average weekly returns on delta-hedged positions, based on market prices at the option expiration day in a month and the position's payoff at the option maturity day in the next month. Delta-hedged returns and factor alphas are computed as in Table 2. The last column shows value-weighted returns/alphas on the 10 – 1 portfolio, where the weighting is according to the market value of the option's open interest (opt-vw). All returns and alphas are in percent per week and the corresponding *t*-statistics are in parentheses. The sample period is 1996-2020.

Table 14

Delta-hedged option portfolio returns for S&P 500 stocks and for stocks ranked 1001–2000 by size capitalization

	1	2	3	4	5	6	7	8	9	10	10–1	10–1 opt-vw
A. Delta-hedged calls, S&P 500 stocks												
Mean excess return	–0.16 (–2.51)	–0.04 (–0.59)	–0.12 (–2.64)	–0.15 (–3.40)	–0.13 (–2.63)	–0.12 (–2.54)	–0.14 (–3.63)	–0.12 (–2.82)	–0.15 (–3.79)	–0.13 (–3.41)	0.03 (0.59)	–0.02 (–0.26)
7-factor alpha	–0.07 (–1.57)	–0.04 (–0.84)	–0.06 (–2.34)	–0.10 (–3.86)	–0.07 (–2.45)	–0.09 (–2.60)	–0.10 (–4.57)	–0.08 (–3.51)	–0.12 (–4.97)	–0.11 (–4.43)	–0.04 (–0.82)	–0.10 (–1.47)
B. Delta-hedged puts, S&P 500 stocks												
Mean excess return	–0.09 (–1.64)	–0.02 (–0.32)	–0.08 (–1.89)	–0.11 (–2.78)	–0.11 (–2.81)	–0.11 (–2.83)	–0.12 (–3.23)	–0.10 (–2.66)	–0.11 (–2.98)	–0.08 (–2.22)	0.01 (0.21)	0.07 (1.22)
7-factor alpha	–0.01 (–0.34)	–0.02 (–0.58)	–0.04 (–1.49)	–0.06 (–2.72)	–0.08 (–3.57)	–0.10 (–4.17)	–0.09 (–4.22)	–0.07 (–3.44)	–0.08 (–3.95)	–0.06 (–2.80)	–0.05 (–1.19)	0.02 (0.33)
C. Delta-hedged calls, stocks ranked 1001–2000												
Mean excess return	–1.02 (–12.74)	–0.62 (–8.62)	–0.45 (–7.00)	–0.30 (–4.63)	–0.24 (–3.96)	–0.26 (–4.42)	–0.20 (–3.48)	–0.11 (–1.59)	–0.14 (–2.25)	–0.07 (–1.06)	0.94 (12.50)	0.90 (6.69)
7-factor alpha	–0.92 (–14.10)	–0.56 (–10.03)	–0.36 (–7.28)	–0.21 (–4.57)	–0.17 (–3.98)	–0.18 (–4.27)	–0.10 (–2.50)	–0.10 (–1.70)	–0.10 (–1.91)	0.03 (0.45)	0.95 (12.27)	0.94 (6.63)
D. Delta-hedged puts, stocks ranked 1001–2000												
Mean excess return	–0.60 (–10.58)	–0.43 (–7.33)	–0.28 (–5.05)	–0.20 (–3.47)	–0.20 (–3.86)	–0.21 (–4.15)	–0.17 (–3.42)	–0.09 (–1.30)	–0.10 (–1.80)	–0.09 (–1.79)	0.51 (10.08)	0.43 (4.16)
7-factor alpha	–0.57 (–11.75)	–0.39 (–8.02)	–0.21 (–4.90)	–0.15 (–3.84)	–0.15 (–3.90)	–0.13 (–3.64)	–0.09 (–2.61)	–0.08 (–1.40)	–0.07 (–1.54)	–0.05 (–1.32)	0.52 (9.84)	0.50 (4.62)

Portfolios are sorted as in Table 1 (decile 1 = lowest stock price, decile 10 = highest stock price) except that we use a different sample of stocks. For each option we calculate the average weekly returns on delta-hedged positions, based on market prices at the option expiration day in a month and the position's payoff at the option maturity day in the next month. Delta-hedged returns and factor alphas are computed as in Table 2. We show results separately for S&P 500 stocks (panels A and B) and for stocks ranked 1001–2000 by market capitalization (panels C and D). All returns and alphas are in percent per week, and the corresponding *t*-statistics are in parentheses. The sample period is 1996–2020.

in investor composition may help explain the weaker effect observed in options on S&P 500 stocks.

The double-sorted portfolio results in [Table 3](#) provide additional insight into the role of liquidity. We find no significant relationship between the underlying stock price and option returns for the most liquid stocks (bottom quintile of stock illiquidity), with a statistically insignificant 7-factor alpha of -0.03% per week for delta-hedged calls. However, the effect is strong in other illiquidity quintiles, particularly for call options, reaching 1.32% per week (t -statistic = 8.60) in the highest illiquidity quintile. When examining option-specific liquidity measures, we find fairly uniform results across option volume quintiles, with 7-factor alphas ranging from 0.52% to 0.64% per week for calls and 0.30% to 0.38% for puts, all statistically significant at the 1% level. The effect varies with percentage bid-ask spreads, being stronger for options with high spreads (0.93% per week for the highest quintile vs. 0.26% for the lowest quintile for calls, and 0.47% vs. 0.15% for puts). However, when examining dollar bid-ask spreads, we find more uniform results, with factor alphas for the 10–1 portfolios ranging between 0.49% and 0.67% for calls and between 0.34% and 0.40% for puts, all maintaining strong statistical significance.

In addition, we control for stock and option illiquidity in Fama-MacBeth regressions reported in [Table 4](#). Indeed, in that table, the coefficients on stock illiquidity and option bid-ask spread are negative and significant, which suggests the importance of illiquidity in shaping the effect of underlying stock price on option expensiveness. However, the coefficients on stock price remain positive and statistically significant after controlling for stock and option illiquidity measures.

In summary, while our results are generally stronger for less liquid stocks, they are not fully explained by liquidity considerations. The relationship between underlying stock prices and option returns disappears in some portfolios of highly liquid securities—specifically, options on S&P 500 stocks and stocks in the bottom illiquidity quintile. However, it remains significant even after controlling for various dimensions of liquidity in both the stock and options markets. This suggests that while limited liquidity may impede arbitrageurs from eliminating the relative expensiveness of options on low-price stocks, it is unlikely to fully explain our findings.

Pick-off risk

A potential alternative mechanism capable of generating return patterns that we document is pick-off risk. Generally, pick-off risk arises when a limit order is filled at a price that is no longer attractive to the trader due to changes in the asset value and the trader's inability to continuously monitor it. Many articles examine the trade-off between pick-off risk, execution probabilities, and execution prices in theoretical and empirical settings. For example, [Foucault \(1999\)](#) builds a dynamic model with sequentially arriving traders who optimally choose between market

and limit orders in the presence of pick-off risk and derives empirical predictions about the proportion of limit and market orders. Hollifield, Miller, and Sandås (2004) derive the properties of the order submission strategy and test their predictions using data from the Stockholm Stock Exchange.¹⁵

In option markets, pick-off risk arises due to market makers' potential inability to react instantaneously to changes in the underlying price, creating some degree of staleness in the bid-ask quotes. Because option prices are convex in the underlying price, market makers might set the midpoint bid-ask quotes above the fundamental option value. Furthermore, pick-off risk is likely higher for options on low-price stocks, because these stocks are more volatile and less liquid. It is therefore possible that pick-off risk alone creates a seeming expensiveness of options on low-price stocks relative to those on high-price stocks.

The results in Table 13 provide some evidence supporting the relevance of pick-off risk. When we exclude options with bid prices below \$0.40—where the potential divergence between midpoint quotes and fair option values is likely to be most severe—our results weaken but remain economically and statistically significant. For example, the 7-factor alpha for the delta-hedged call portfolio decreases from 0.58% to 0.38% per week.

However, we believe that pick-off risk, by itself, is unable to explain the totality of our evidence. First, as we discuss in Section Liquidity tests, our results, although affected by illiquidity measures, generally remain statistically significant after controlling for illiquidity. Second, if pick-off risk is the only mechanism driving our results, retail demand should be irrelevant and should not affect option prices. However, as we show in Table 6, the expensiveness of options on low-price stocks is positively and significantly related to retail demand. Finally, in Table 12, we show that the relative expensiveness of options on low-price stocks remains significant for calls even when the effective bid-ask spread is equal to 50% of the actual spread (for puts, the results are significant at a 25% effective spread). If the relation between stock price and option expensiveness is driven solely by pick-off risk, this evidence implies that, at least for call options, market makers are forced to move the midpoint quotes relative to the fundamental value by more than 50% of the actual bid-ask spreads, a number that appears unrealistically high.

In summary, there is evidence that liquidity, limits to arbitrage, and strategic quote setting by option market makers contribute to the effect of underlying stock price on option expensiveness. However, our results remain significant (albeit weaker in magnitude) in subsamples where limits to arbitrage and pick-off risk are likely less severe, and after controlling for various illiquidity measures. This points back to the role of retail investors coupled with divergence of opinion and skewness preference as factors driving our results.

¹⁵ See also Cohen et al. (1981), Harris (1998), and Parlour (1998) for models of optimal placement strategies with market and limit orders.

Conclusion

We document a new effect in options prices; that is, options on low-price stocks tend to be relatively expensive. This expensiveness is manifested in delta-hedged call and put portfolio returns, in returns to portfolios double sorted on the underlying price and various stock characteristics, as well as in cross-sectional FM regressions. Our results are robust to using an alternative measure of mispricing based on the difference between market and theoretical Black-Scholes option values.

We argue that budget-constrained retail investors cause this expensiveness of options on low-price stocks. We present evidence that retail investors are net buyers of options on low-price stocks and that option expensiveness is stronger for options traded more heavily by retail traders. A series of natural experiments corroborates our conclusion; we find the relative expensiveness of mini contract options, options postsplit, and also of mini index options (relative to main index options).

We explore two potential channels that can result in option expensiveness: retail investors' preference for skewness and divergence of opinion. We find evidence in support of both mechanisms. We also examine the role of liquidity, limits to arbitrage, and strategic quote setting by market makers and find that those factors also likely contribute to our findings. However, the persistence of the results in portfolios of more liquid stocks and after controlling for various trading frictions suggests that retail investor behavior—driven by skewness preferences and divergence of opinion—plays an important role in the observed expensiveness of options on low-price stocks.

Code Availability

The replication code is available in the Harvard Dataverse at <https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/9G30TK>

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